3FFA/3FFA4 Fourier and Functional Analysis

Lecture notes

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Questions and comments:

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§1 Vector spaces

Throughout the course we shall be using the notion of a *vector space*.

Definition (Vector space)

Let X be a non-empty set. We say X is a *vector space* (or *vector space over* $\mathbb{R}$) if there are two binary operations
$+ : X \times X \to X$ and $\times : \mathbb{R} \times X \to X$ such that $(X, +)$ is an Abelian group and the following 4 further axioms are satisfied:

- $\alpha(x + y) = \alpha x + \alpha y$ for all scalars $\alpha \in \mathbb{R}$ and $x, y \in X$;
- $(\alpha + \beta)x = \alpha x + \beta x$ for all scalars $\alpha, \beta \in \mathbb{R}$ and $x \in X$;
- $\alpha(\beta x) = (\alpha\beta)x$ for all scalars $\alpha, \beta \in \mathbb{R}$ and $x \in X$;
- $1 \cdot x = x$ for all $x \in X$.

Remark. If $\mathbb{R}$ is replaced by $\mathbb{C}$ in the definition above, we get a *vector space over* $\mathbb{C}$.

Example (Vector spaces)

1. For $n \in \mathbb{N}$, $\mathbb{R}^n = \{(a_1, \dots, a_n) : a_i \in \mathbb{R}\}$ is a vector space over $\mathbb{R}$. Zero $\mathbf{0}_{\mathbb{R}^n} = (0, \dots, 0)$.
2. Let $L = \{(a_1, a_2, \dots) = (a_k)_{k \geqslant 1} : a_k \in \mathbb{R} \text{ for all } k \geqslant 1\}$. Then L is a vector space over $\mathbb{R}$ if we define addition and scalar multiplication in the usual way: for $(a_k)_{k\geqslant 1}, (b_k)_{k \geqslant 1} \in L$ and $\alpha \in \mathbb{R}$
   $(a_k)_{k\geqslant 1} + (b_k)_{k\geqslant 1} := (a_k + b_k)_{k \geqslant 1}$, $\alpha \times (a_k)_{k\geqslant 1} := (\alpha a_k)_{k\geqslant 1}$, zero $\mathbf{0}_L = (0, 0, \dots)$.
3. Let $a < b$ be real numbers. Define

   $$C[a, b] := \{f : [a, b] \to \mathbb{R} : f \text{ is continuous}\}$$

   to be the set of all continuous real-valued functions on the interval $[a, b] \subseteq \mathbb{R}$. Then $C[a, b]$ is a vector space over $\mathbb{R}$ if we define addition and scalar multiplication in the following way: for each $t \in [a, b]$, let for any $f, g \in C[a, b]$ and $\alpha \in \mathbb{R}$
   $(f+g)(t) := f(t) + g(t)$, $(\alpha f)(t) := \alpha f(t)$, zero $\mathbf{0}_{C[a,b]}$ is the constant zero function. Of course, sum of two continuous functions is a continuous function, and if we multiply a continuous function by a scalar, we get a continuous function.

Recall, if X is a vector space and $\emptyset \neq Y \subseteq X$, then Y is called a *linear subspace* of X if it is a vector space with respect to addition and scalar multiplication defined on X. In particular, $\mathbf{0}_X$ must belong to Y.

Example (Linear subspaces)

1. Linear subspaces of $\mathbb{R}^n$: $\{(x_1, x_2, \dots, x_{n-1}, 0) : x_i \in \mathbb{R}\}$, $\{(x, x, 0, \dots, 0) : x \in \mathbb{R}\}$, $\{(x, 0, y, 0, \dots, 0) : x, y \in \mathbb{R}\}$. All these contain $\mathbf{0}_{\mathbb{R}^n}$.

   In your notes, write in two or three more examples not listed above.

2. Linear subspaces of L:
   $\triangleright$ Any $\mathbb{R}^n$ for some $n \geqslant 1$ is a linear subspace of L. We identify $\mathbb{R}^n$ with
   $L_n = \{(a_k)_{k\geqslant 1} : a_k \in \mathbb{R}\, \forall k \geqslant 1,\ \ a_k = 0\, \forall k \geqslant n+1\}$
   $= \{(a_1, \dots, a_n, 0, 0, 0, 0, \dots) : a_k \in \mathbb{R} \text{ for } 1 \leqslant k \leqslant n\}$. Zero $\mathbf{0}_L \in L_n$.
   $\triangleright$ $c_0 = \{(a_k) \mid a_k \to 0\}$. Zero $\mathbf{0}_L \in c_0$.
   $\triangleright$ $c_{00} = \{(a_k) \mid \exists N \geqslant 1 \text{ s.t. } a_k = 0 \quad \forall k \geqslant N\}$. Zero $\mathbf{0}_L \in c_{00}$.

   Lemma 1.1. *For every $n \geqslant 1$, $c_{00} \neq L_n$.*

   Proof. Let $n \geqslant 1$ be fixed. Then $v_n = (\underbrace{0, \dots, 0}_{n \text{ times}}, 1, 0, 0, 0, \dots) \in c_{00} \setminus L_n$. □

3. One of linear subspaces of $C[a, b]$ is $\mathcal{P}[a, b] = \{p|_{[a,b]} \,\big|\, p : \mathbb{R} \to \mathbb{R} \text{ is a polynomial}\}$. Zero $\mathbf{0}_{C[a,b]} \in \mathcal{P}[a, b]$.

   Note that if $a < b$, then for two distinct polynomials $p_1 \neq p_2$, their restrictions to $[a, b]$ are distinct elements of $C[a, b]$: if $p_1(x) = p_2(x)$ for all $x \in [a, b]$, then the polynomial $p_1 - p_2$ has infinitely many roots, so must be identically zero.

QUICK POLLS: 1. VECTOR SPACES See PowerPoint presentation 1, questions 1-2.

Remark. Note that in the last example all members of the collection $\mathcal{P}[a, b]$ can be expressed as a **finite** sum of scalar multiples of "basic" functions $1, t, t^2, t^3, \dots$. This brings us to the following definition.

Definition (Linear span)

If M is a subset of a vector space X then

$$\langle M \rangle = \{\sum_{j=1}^{n} \alpha_j v_j : v_j \in M, \alpha_j \in \mathbb{R}\}$$

is called the *linear span* of M.

Example (Linear span.)

1. $X = \mathbb{R}^n$. Define vectors $e_1 = (1, \underbrace{0, 0, \dots, 0}_{n-1 \text{ times}})$, $e_2 = (0, 1, \underbrace{0, \dots, 0}_{n-2 \text{ times}}), \dots, e_n = (\underbrace{0, 0, \dots, 0}_{n-1 \text{ times}}, 1)$; let $M = \{e_1, e_2, \dots, e_n\}$.

Then $\langle M \rangle = \mathbb{R}^n$.

2. Similarly, as in the previous example, for each $n \geqslant 1$, define vectors
$e_n = (\underbrace{0, 0, \ldots, 0}_{n-1 \text{ times}}, 1, 0, 0, \ldots) \in L$.
Let $M = \{e_k : k \geqslant 1\}$. What is $\langle M \rangle$?
QUICK POLLS: 1. VECTOR SPACES See PowerPoint presentation 1, question 3.

Remark. If X is a vector space then we say it is *finite-dimensional* if there is a finite subset M of X such that $\langle M \rangle = X$.
If X is not finite-dimensional, it is called *infinite-dimensional*.

Example (Finite- and infinite-dimensional vector spaces)

1. Space $\mathbb{R}^n$ is finite-dimensional, take M to be the collection of n vectors $e_1, \ldots, e_n$, where e_k has kth coordinate 1 and all other coordinates 0 (first example of a linear span above).
2. L is an infinite-dimensional space.
3. $C[a, b]$ is an infinite-dimensional space.

We will prove that L and $C[a, b]$ are infinite-dimensional later. See Remark after Corollary 2.16.

§2 Normed spaces

Definition (Norm)

Let X be a vector space over $\mathbb{R}$. A *norm* on X is a map $\|\cdot\| : X \to \mathbb{R}$ such that

1. $\|x\| \geqslant 0$ for all $x \in X$, and if $\|x\| = 0$ then $x = 0$;
2. $\|\alpha x\| = |\alpha|\|x\|$ for all $\alpha \in \mathbb{R}$, $x \in X$;
3. $\|x + y\| \leqslant \|x\| + \|y\|$ for all $x, y \in X$.

Then $(X, \|\cdot\|)$ is a *normed space*.

Remark. 1. In the definition of the norm we require that $\|x\| = 0$ implies $x = 0$. However, the other implication is always satisfied. Indeed, $\|0\| = \|0 \cdot 0\| = |0|\|0\| = 0$. This follows from the 2nd property of the norm.

2. We will often refer to the third property of the norm, $\|x + y\| \leqslant \|x\| + \|y\|$, as the *triangle inequality*.

Lemma 2.1. *Let $(X, \|\cdot\|)$ be a normed space and $Y \subseteq X$ its linear subspace. Then the formula $\|y\|_Y := \|y\|$ defines a norm on Y.*

Proof. All three properties from the definition of the norm are satisfied. □

Remark. If $(X, \|\cdot\|)$ is a normed space, then $d(x, y) = \|x - y\|$ defines a metric on X.

Examples 2.2 (Normed spaces)**.** The following functions are norms.

(1) For $p \in [1, \infty]$, the pairs $(\mathbb{R}^n, \|\cdot\|_p)$ and $(\mathbb{C}^n, \|\cdot\|_p)$ are normed vector spaces, where for $x = (a_1, ..., a_n)$

$$\begin{cases} \|x\|_p = \|(a_1, \dots, a_n)\|_p = \big(\sum\limits_{k=1}^{n} |a_k|^p\big)^{1/p} & \text{for } p \in [1, \infty), \\ \|x\|_\infty = \|(a_1, \dots, a_n)\|_\infty = \max\Big\{|a_k| : k = 1, \dots, n\Big\} & \text{for } p = \infty. \end{cases} \qquad (*)$$

1a: if $p = 2$ then this is our usual distance in $\mathbb{R}^n$: $\|(a_1, \dots, a_n)\|_2 = \sqrt{|a_1|^2 + \dots + |a_n|^2}$.

(2) If Y is a finite-dimensional space over $\mathbb{F}$ and $\{e_1, \dots, e_n\}$ is a basis of Y, then for each $p \in [1, +\infty]$ the following formula defines a norm on Y:

$$y \in Y \quad \mapsto \quad y = \sum_{i=1}^{n} a_i e_i \quad \mapsto \quad \|y\|_p = \|(a_1, \dots, a_n)\|_p,$$

where the latter uses the formula $(*)$ above.

(3) For $p \in [1,\infty)$, let ℓ^p denote the set of all infinite sequences $x = (a_k)_{k\geqslant 1}$, where $a_k \in \mathbb{R}$ for each $k \in \mathbb{N}$, such that $\sum_{k\geqslant 1} |a_k|^p < \infty$. If $\|x\|_p := \big(\sum_{k=1}^{\infty} |a_k|^p\big)^{1/p}$ then $(\ell^p, \|\cdot\|_p)$ is a normed vector space.

For $p = \infty$ we let ℓ^∞ denote the set of all bounded sequences $x = (a_k)_{k\geqslant 1}$, where $a_k \in \mathbb{R}$, and for $x \in \ell^\infty$ define $\|x\|_\infty := \sup\big\{|a_k| : k \geqslant 1\big\}$. Then $(l^\infty, \|\cdot\|_\infty)$ is a normed vector space. We shall prove this later.

(4) For real numbers $a < b$, and we define the following different norms on the same vector space $C[a,b]$:

$$\|f\|_1 := \int_a^b |f(t)|dt, \qquad \text{integral norm;}$$
$$\|f\|_\infty := \sup\{|f(t)| : t \in [a,b]\}, \qquad \text{sup-norm.}$$

Remark. Note that in each of the examples above, the value of $\|x\|_p$ or $\|f\|_p$ is well-defined:

(1) Here we are talking about finite sums and a maximum of finitely many numbers, this is always finite;

(2) Here $(a_1,\dots,a_n) \in \mathbb{R}^n$, so its ℓ^p-norm is well defined by (1) above.

(3) Here we defined spaces (collections) ℓ^p in such a way that $\|x\|_p$ is necessarily finite, and we may also use the same formula to define ℓ^p over $\mathbb{C}$.

(4) Here we use that if f is continuous, then $|f|$ is also continuous, hence $|f|$ is bounded (so that sup-norm of f is well-defined) and $|f|$ is Riemann integrable (so that the integral norm of f is well-defined).

QUICK POLLS: 2. NORMED SPACES See PowerPoint presentation 2, question 1.

Proposition 2.3 (Minkowski's inequality)**.** *Let $x = (x_k)_{k\geqslant 1}$ and $y = (y_k)_{k\geqslant 1}$ be two elements of ℓ^p, where $p \in [1,\infty]$. Then the sequence $x + y = (x_k + y_k)_{k\geqslant 1}$ belongs to ℓ^p and*

$$\|x+y\|_p \leqslant \|x\|_p + \|y\|_p. \tag{2.1}$$

Proof. Let $x = (x_k)_{k\geqslant 1}$ and $y = (y_k)_{k\geqslant 1}$.

Case 1: When $p = \infty$, we have

$$|x_k + y_k| \leqslant |x_k| + |y_k| \leqslant \|x\|_\infty + \|y\|_\infty = \text{Const} < \infty$$

for each $k \in \mathbb{N}$ and therefore $x + y$ is a bounded sequence (belongs to ℓ^∞) and $\|x+y\|_\infty = \sup_{k\in\mathbb{N}} |x_k + y_k| \leqslant \|x\|_\infty + \|y\|_\infty$.

Case 2: Suppose now that $p \in [1,\infty)$. We first remark that if $\|x\|_p = 0$ then all $x_k = 0$ and so $x + y = y$ and the result holds.

We may assume therefore $\|x\|_p$ and $\|y\|_p$ are both nonzero.

Define

$$a_k := \frac{|x_k|}{\|x\|_p}, \quad b_k := \frac{|y_k|}{\|y\|_p}$$

for each $k \in \mathbb{N}$, then $a_k, b_k \geqslant 0$ for each k,

$$\sum_{k\in\mathbb{N}} a_k^p = \big(\sum_{k\in\mathbb{N}} |x_k|^p\big)/\|x\|_p^p = 1 \quad \text{and}$$
$$\sum_{k\in\mathbb{N}} b_k^p = \big(\sum_{k\in\mathbb{N}} |y_k|^p\big)/\|y\|_p^p = 1.$$

Fix any $k \in \mathbb{N}$; then we have:

$$\begin{aligned} |x_k + y_k|^p &\leqslant (|x_k| + |y_k|)^p \\ &= (\|x\|_p a_k + \|y\|_p b_k)^p \\ &= (\|x\|_p + \|y\|_p)^p \Big(\frac{\|x\|_p}{\|x\|_p + \|y\|_p} a_k + \frac{\|y\|_p}{\|x\|_p + \|y\|_p} b_k\Big)^p \\ &= (\|x\|_p + \|y\|_p)^p (\alpha a_k + (1-\alpha) b_k)^p, \end{aligned}$$

where $\alpha = \|x\|_p/(\|x\|_p + \|y\|_p) \in (0,1)$. Since the function $H(t) = t^p : [0,\infty) \to [0,\infty)$ is *convex* for $p \geqslant 1$ it follows that

$$H(\alpha a_k + (1-\alpha) b_k) \leqslant \alpha H(a_k) + (1-\alpha) H(b_k)$$

and so

$$|x_k + y_k|^p \leqslant (\|x\|_p + \|y\|_p)^p (\alpha a_k^p + (1-\alpha) b_k^p).$$

Taking the sum over all $k \geqslant 1$ it follows that

$$\begin{aligned} \sum_{k\geqslant 1} |x_k + y_k|^p &\leqslant (\|x\|_p + \|y\|_p)^p (\alpha \sum_{k\geqslant 1} a_k^p + (1-\alpha) \sum_{k\geqslant 1} b_k^p) \\ &= (\|x\|_p + \|y\|_p)^p (\alpha \cdot 1 + (1-\alpha)\cdot 1) = (\|x\|_p + \|y\|_p)^p. \end{aligned}$$

Hence $\sum_{k\geqslant 1} |x_k + y_k|^p$ converges and $\Big(\sum_{k\geqslant 1} |x_k + y_k|^p\Big)^{1/p} \leqslant \|x\|_p + \|y\|_p$ as required. □

Corollary 2.4. *For each $1 \leqslant p \leqslant \infty$ we have that*
- ℓ^p is a vector space, and
- $(\ell^p, \|\cdot\|_p)$ is a normed space.

Proof. 1. Let us show first that ℓ^p is a vector space. It is a subset of a vector space $\mathcal{L}$ of all sequences, so it is enough to check:

- $0 \in \ell^p$: this is true as $(0,0,\dots) \in \ell^p$, $\sum |0|^p < \infty$ and $\sup |0| < \infty$;
- If $x, y \in \ell^p$, then $x + y \in \ell^p$ by Proposition 2.3;
- It is easy to check that for $c \in \mathbb{R}$ and $x \in \ell^p$ the vector (sequence) cx also belongs to ℓ^p.

2. $(\ell^p, \|\cdot\|_p)$ is a normed space: For any $x \in \ell^p$ we have $\|x\|_p$ is finite by definition of $\|\cdot\|_p$ and ℓ^p itself. Now let us verify the three properties that the norm must satisfy:

(i) $\|x\|_p \geqslant 0$ as definition of $\|x\|_p$ for both $1 \leqslant p < \infty$ and $p = \infty$ is based on $|a_k| \geqslant 0$; if $\|x\|_p = 0$ then all $a_k = 0$, i.e. $x = (0, 0, 0, \dots)$ is the zero vector – this was proved in Proposition 2.3;

(ii) $\|cx\|_p = |c|\|x\|_p$ for all $c \in \mathbb{R}$ and $x \in \ell^p$;

(iii) Triangle inequality for $\|\cdot\|_p$ follows from Minkowski's inequality, Proposition 2.3. □

Important and interesting facts about subspaces of ℓ^p spaces.

1. ℓ^p are vector spaces (proved in Corollary 2.4) and they are all different:
If $1 \leqslant p < q \leqslant \infty$, then $\ell^p \subseteq \ell^q$ and $\ell^p \neq \ell^q$, see Formative Problem Sheet 1.

By Lemma 2.1, this implies that ℓ^p is a linear subspace of ℓ^q and the norm $\|\cdot\|_q$ is well-defined on ℓ^p.

2. $L_n \subsetneq c_{00} \subsetneq \ell^p$ for every $n \geqslant 1$ and every $1 \leqslant p \leqslant \infty$ since $\|x\|_p$ is finite whenever x has only finitely many non-zero terms. Hence L_n and c_{00} are linear subspaces of ℓ^p, for every $1 \leqslant p \leqslant \infty$.

QUICK POLLS: 2. NORMED SPACES See PowerPoint presentation 2, question 2.

Corollary 2.5. *For all $1 \leqslant p \leqslant \infty$ and integer $n \geqslant 1$ the finite-dimensional $(\mathbb{R}^n, \|\cdot\|_p)$ is a normed space. Moreover, if Y is a finite-dimentional space and $\{e_1, \dots, e_n\}$ is a basis of Y, then $N(\sum_{i=1}^n a_i e_i) = \|(a_1, \dots, a_n)\|_p$ defines a norm on Y.*

Proof. Since L_n is a subset of ℓ^p and a vector space, it is a linear subspace of ℓ^p. By Lemma 2.1 $(L_n, \|\cdot\|_p)$ is a normed space.

To see that $(\mathbb{R}^n, \|\cdot\|_p)$ is a normed space, identify $\mathbb{R}^n$ with L_n via $(a_1, \dots, a_n) \mapsto (a_1, \dots, a_n, 0, 0, 0, 0, 0, \dots)$, and note that addition and scalar multiplication on L_n keep this correspondence intact. The values of $\|\cdot\|_p$ on corresponding vectors from $\mathbb{R}^n$ and L_n are the same.

To justify the "moreover" part about Y, note that for each y there exists a unique n-tuple $(a_1, \dots, a_n) \in \mathbb{R}^n$ such that $y = \sum_{i=1}^n a_i e_i$, hence $N(y)$ is well-defined. Since vector addition and scalar multiplication on Y correspond to the same operations on $\mathbb{R}^n$, the function $N(\cdot)$ satisfies all properties of a norm on Y. □

Remark. We now see that Examples 2.2, parts (1) and (2), of normed spaces follow from Corollary 2.5, and part (3) follows from Corollary 2.4. The last part (4) will be proved in the following Theorem 2.6 and Question 3 from the Problem Sheet 1.

Theorem 2.6. *$(C[a, b], \|\cdot\|_1)$ and $(C[a, b], \|\cdot\|_\infty)$ are normed spaces.*

Proof. $(C[a, b], \|\cdot\|_1)$ - see Problem Sheet 1, Question 3(a).

Note that for any $f \in C[a, b]$ there is t_0 such that $\|f\|_\infty = |f(t_0)|$. Hence $\|f\|_\infty$ is a finite number for any $f \in C[a, b]$. We now verify the three properties of the norm.

1. It is clear that $\|f\|_\infty \geqslant 0$ for any $f \in C[a,b]$.

   If $\|f\|_\infty = 0$ then $|f(t)| = 0$ for all $t \in [a,b]$ thus $f(t) = 0$ for all $t \in [a,b]$, i.e., $f = 0$.

2. Let $c \in \mathbb{R}$, then $\sup\{|cf(t)| : t \in [a,b]\} = |c| \sup\{|f(t)| : t \in [a,b]\} = |c|\|f\|_\infty$ as $|c| \geqslant 0$.

3. Let $f, g \in C[a,b]$, then $\|f+g\|_\infty = \sup\{|f(t)+g(t)| : t \in [a,b]\}$.

   Denote $A = \|f\|_\infty$ and $B = \|g\|_\infty$. Then $|f(t)| \leqslant A$ and $|g(t)| \leqslant B$ for all $t \in [a,b]$. Thus $|f(t)+g(t)| \leqslant |f(t)| + |g(t)| \leqslant A + B$ for every $t \in [a,b]$. This implies

$$\|f+g\|_\infty = \sup\Big\{|f(t)+g(t)| : t \in [a,b]\Big\} \leqslant A + B = \|f\|_\infty + \|g\|_\infty.$$

□

Convergence in normed spaces

We conclude this section by introducing the notion of convergence in normed vector spaces.

Definition (Converging & diverging sequence)

Let $(X, \|\cdot\|)$ be a normed vector space and let $v_n \in X$ be a sequence of vectors in X. If $v \in X$ is such that

$$\|v - v_n\| \underset{n\to\infty}{\to} 0$$

then (v_n) is said to *converge* to v, which we write as $v_n \underset{n\to\infty}{\to} v$ and $\lim v_n = v$.

Remark. Notice that $\|v_n - v\| = \|-1 \cdot (v - v_n)\| = |-1| \cdot \|v - v_n\| = \|v - v_n\|$, so when it is more convenient, we can calculate $\|v_n - v\|$ instead of $\|v - v_n\|$.

Examples 2.7 (Convergence in normed spaces)**.**
(1) If $v = (a_k) \in \ell^1$ and $v_n = (a_1, \dots, a_n, 0, 0, 0, \dots)$, then $v_n \to v$ in $(\ell^1, \|\cdot\|_1)$.

Proof. Let us calculate $\|v - v_n\|_1$. We have $\|v - v_n\|_1 = \big\|(a_1, \dots, a_n, a_{n+1}, a_{n+2}, \dots) - (a_1, \dots, a_n, 0, 0, 0, \dots)\big\|_1 = \|(0, 0, \dots, 0, a_{n+1}, a_{n+2}, \dots)\|_1 = \sum_{k=n+1}^{\infty} |a_k|$. Note that as $\sum_{k=1}^{\infty} |a_k|$ converges, we have that $\|v - v_n\|_1 = \sum_{k=n+1}^{\infty} |a_k| \to 0$ as $n \to \infty$. □

(2–3) QUICK POLLS: 2. NORMED SPACES See PowerPoint presentation 2, questions 3,4.

(2): $v_n = (\underbrace{\frac{1}{n}, \dots, \frac{1}{n}}_{n \text{ times}}, 0, 0, \dots)$; (3) $u_n = (\underbrace{1, \dots, 1}_{n \text{ times}}, 0, 0, \dots)$.

Theorem 2.8. *Let $(X, \|\cdot\|)$ be a normed vector space.*

(1) If (v_n) is a convergent sequence in X with limits v and u, then $v = u$ (uniqueness of limits).

(2) If (v_n) *and* (u_n) *are convergent sequences in* X *with limits* v *and* u *respectively then* $(v_n + u_n)$ *is a convergent sequence in* X *with limit* $v + u$.

(3) If (α_n) *is a convergent sequence in* $\mathbb{R}$ *and* (v_n) *is a convergent sequence in* X *with limits* α *and* v *respectively, then* $(\alpha_n v_n)$ *is a convergent sequence in* X *with limit* αv.

(4) If (v_n) *is a convergent sequence in* X *with limit* v *then* $(\|v_n\|)$ *is a convergent sequence in* $[0, \infty)$ *with limit* $\|v\|$.

Proof. (1) Note that $0 \leqslant \|v - u\| = \|v - v_n + v_n - u\| \leqslant \|v - v_n\| + \|v_n - u\| \to 0$ as $v_n \to v$ and $v_n \to u$.

As $\|v - u\|$ is a fixed number independent of n, this may only happen if $\|v - u\| = 0$.

By the first property of a norm, this implies $v - u = 0$, i.e. $v = u$.

(2) We use $0 \leqslant \|(v+u) - (v_n + u_n)\| \leqslant \|v - v_n\| + \|u - u_n\| \to 0$ to show $(v_n + u_n) \to (v + u)$.

(3) By the triangle inequality, $0 \leqslant \|\alpha_n v_n - \alpha v\| = \|\alpha_n(v_n - v) + (\alpha_n - \alpha)v\|$
$\leqslant |\alpha_n|\|v_n - v\| + |\alpha_n - \alpha|\|v\| \underset{n\to\infty}{\to} |\alpha| \cdot 0 + 0 \cdot \|v\| = 0$. Hence $\|\alpha_n v_n - \alpha v\| \to 0$. By definition this means $\alpha_n v_n \to \alpha v$.

(4) Using $0 \leqslant \Big|\|v_n\| - \|v\|\Big| \leqslant \|v_n - v\| \to 0$ (from Exercise 2(a), Problem sheet 1) we get the required. □

Definition (Cauchy sequence)

Let $(X, \|\cdot\|)$ be a normed vector space and let (v_n) be a sequence of vectors in X. Then (v_n) is said to be a *Cauchy sequence* in $(X, \|\cdot\|)$ if

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n, m \geqslant N \quad (\|v_n - v_m\| < \varepsilon). \tag{2.2}$$

Proposition 2.9. *Let* $(X, \|\cdot\|)$ *be a normed vector space and let* (v_n) *be a convergent sequence in* X. *Then* (v_n) *is Cauchy.*

Proof. Let $v_n \to v$ in $(X, \|\cdot\|)$. Fix $\varepsilon > 0$. By definition of convergence there exists N such that if $k \geqslant N$, then $\|v_k - v\| < \varepsilon/2$.

Let $n, m \geqslant N$. By triangle inequality we have $\|v_n - v_m\| \leqslant \|v_n - v\| + \|v - v_m\| = \|v_n - v\| + \|v_m - v\| < \varepsilon$. Hence (v_n) is Cauchy. □

Examples 2.7 (2), (3) – revisited

Getting back to Example (3) above, where $(X, \|\cdot\|) = (\ell^\infty, \|\cdot\|_\infty)$ and $u_n = (\underbrace{1, 1, \dots, 1}_{n \text{ times}}, 0, 0, \dots)$ we notice that for all $n \neq m$ we have (assuming $n > m$) $u_n - u_m = (\underbrace{0, \dots, 0}_{m \text{ times}}, \underbrace{1, \dots, 1}_{n-m \text{ times}}, 0, 0, \dots)$, so $\|u_n - u_m\|_\infty = 1$. This means that the sequence (u_n) is not Cauchy, hence it does not

converge at all (by Proposition 2.9). ($\varepsilon = 1/2$ will not work with any N, just take $m = N$ and $n = N + 1$.)

We leave it as an exercise to show that the sequence $v_n = (\underbrace{\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}}_{n \text{ times}}, 0, 0, \dots)$ is not Cauchy in $(\ell^1, \|\cdot\|_1)$, so it does not converge at all too.

Definition (Equivalent norms)

Suppose $\|\cdot\|$ and $|||\cdot|||$ are two norms on the same vector space X. Then these norms are said to be *equivalent* if there exist constants $\alpha, \beta > 0$ such that $\alpha\|x\| \leqslant |||x||| \leqslant \beta\|x\|$ for all $x \in X$. In this case we write $\|\cdot\| \sim |||\cdot|||$.

Remark. The relation $\sim$ is an equivalence relation on the set of norms on X. We prove this in Question 11 of Problem Sheet 1.

Theorem 2.10. *Let X be a vector space. Let $\|\cdot\|$ and $|||\cdot|||$ be two equivalent norms on X. and $(v_n) \subseteq X$ a sequence. Then (v_n) converges in $(X, \|\cdot\|)$ if and only if it converges in $(X, |||\cdot|||)$, and to the same limit.*

Proof. As $\|\cdot\| \sim |||\cdot||| \implies |||\cdot||| \sim \|\cdot\|$, it is enough to prove just one implication.

Assume $\alpha, \beta > 0$ are such that $\alpha\|x\| \leqslant |||x||| \leqslant \beta\|x\|$ for all $x \in X$.

Assume $(v_n) \subseteq X$ converges in $(X, \|\cdot\|)$. Let $x \in X$ be such that $\|v_n - x\| \to 0$.

Then $0 \leqslant |||v_n - x||| \leqslant \beta\|v_n - x\| \to 0$ implies $|||v_n - x||| \to 0$. □

Lemma 2.11. *Let X be a vector space, $\|\cdot\|$ and $|||\cdot|||$ two norms on X. The following conditions are equivalent:*

(1) There is no $C > 0$ such that $\|x\| \leqslant C|||x|||$ for all $x \in X$,

(2) There is a sequence $v_n \in X$ such that $\|v_n\| = 1$ and $|||v_n||| \to 0$.

Proof. (2) $\implies$ (1) Assume such $C > 0$ exists and let $x = v_n$. Then $1 \leqslant C|||v_n||| \to 0$, a contradiction.

(1) $\implies$ (2) For each $C = n \geqslant 1$, let $x_n \in X$ be such that $\|x_n\| > n|||x_n|||$. Then $x_n \neq 0$ for all $n \geqslant 1$. Take $v_n = x_n/\|x_n\|$, then $\|v_n\| = \left\|x_n/\|x_n\|\right\| = \|x_n\|/\|x_n\| = 1$ and $|||v_n||| = |||x_n|||/\|x_n\| < 1/n \to 0$. □

Example 2.7(2) - revisited again

Recall that $\|v_n\|_1 = 1$ and $\|v_n\|_\infty \to 0$. By Lemma 2.11, $\|\cdot\|_\infty$ is not equivalent to $\|\cdot\|_1$ on ℓ^1.

Lemma 2.12. *Let $1 \leqslant p \leqslant +\infty$ and $x = (a_n) \in \ell^p$. Then for each $n \geqslant 1$, we have $|a_n| \leqslant \|x\|_p$.*

Proof. Case 1: $p = +\infty$. Fix any $n \geqslant 1$. Then $\|x\|_\infty = \sup_{k\geqslant 1} |a_k| \geqslant |a_n|$.

Case 2: $1 \leqslant p < +\infty$. For any $n \geqslant 1$, $(\|x\|_p)^p = \sum_{k=1}^\infty |a_k|^p \geqslant |a_n|^p$, hence $\|x\|_p \geqslant |a_n|$. □

Corollary 2.13. *Let $1 \leqslant p \leqslant \infty$. If the sequence $x_m = (a_k^{(m)})_{k\geqslant 1}$ converges (as $m \to \infty$) to $x = (b_k)_{k\geqslant 1}$ in $(\ell^p, \|\cdot\|_p)$, then $a_k^{(m)} \underset{m\to\infty}{\to} b_k$ for each $k \geqslant 1$.*

Proof. This follows from $0 \leqslant |a_k^{(m)} - b_k| \leqslant \|x_n - x\|_p \to 0$. □

Remark. In general, the converse is not true: $\nu_m = (\underbrace{1/m, 1/m, \dots, 1/m}_{m \text{ times}}, 0, 0, 0, \dots)$ does not converge to the zero sequence in $(\ell^1, \|\cdot\|_1)$, see Examples 2.7 and argument in Quick Polls 2. However, this converse **does hold** in **finite-dimensional** spaces.

Remark. Let X be a finite-dimensional space with a basis $\{e_1, \dots, e_n\}$. If the sequences of real numbers converge as follows

$$a_i^{(k)} \underset{k\to\infty}{\to} b_i, \qquad \text{for each } 1 \leqslant i \leqslant n,$$

then $\left\| \sum_{i=1}^n a_i^{(k)} e_i - \sum_{i=1}^n b_i e_i \right\| \to 0$ for any norm $\|\cdot\|$ on X.

In particular, coordinate-wise convergence of vectors in $\mathbb{R}^n$ implies convergence of vectors in $(\mathbb{R}^n, \|\cdot\|_p)$, for every $1 \leqslant p \leqslant \infty$. In other words, in $(\mathbb{R}^n, \|\cdot\|_p)$ a converse to Corollary 2.13 holds.

Proof. This follows from $0 \leqslant \left\| \sum_{i=1}^n a_i^{(k)} e_i - \sum_{i=1}^n b_i e_i \right\| \leqslant \sum_{i=1}^n \left| a_i^{(k)} - b_i \right| \|e_i\| \to \sum_{i=1}^n 0 = 0.$ □

Theorem 2.14. *Let X be a finite-dimensional space and $\|\cdot\|$ and $|||\cdot|||$ be two norms on X. Then $\|\cdot\| \sim |||\cdot|||$.*

Proof. Let $\dim X = n$; let $e_1, \dots, e_n$ be a basis on X and $\|\cdot\|_1$ be the norm defined in Examples 2.2(2). It is enough to show that $\|\cdot\| \sim \|\cdot\|_1$, then from transitivity of equivalence relation (see Problem Sheet 1, Question 11) the statement would follow.

Let $x = \sum_{i=1}^n a_i e_i \in X$ be any, then

$$\|x\| = \left\| \sum_{i=1}^n a_i e_i \right\| \leqslant \sum_{i=1}^n |a_i| \|e_i\| \leqslant \sum_{i=1}^n |a_i| \max_{1\leqslant i\leqslant n} \|e_i\| = \beta \|x\|_1, \text{ with } \beta = \max_{1\leqslant i\leqslant n} \|e_i\| \geqslant \|e_1\| > 0.$$

We now show that there exists $\alpha > 0$ such that $\|x\| \geqslant \alpha \|x\|_1$ for all $x \in X$. Assume this is not true, then by Lemma 2.11 there exists a sequence $\nu_m = \sum_{i=1}^n a_i^{(m)} e_i \in X$ such that $\|\nu_m\| \to 0$ and $1 = \|\nu_m\|_1 = \sum_{i=1}^n |a_i^{(m)}|$, so $|a_i^{(m)}| \leqslant 1$ for each $1 \leqslant i \leqslant n$.

Choose a convergent subsequence $(a_1^{(m_{1,k})})_k$ of $a_1^{(m)}$, with limit b_1; then a convergent subsequence $(a_2^{(m_{2,k})})_k$ of $a_2^{(m_{1,k})}$, with limit b_2; and so on, until we get, on the nth step, a

subsequence $v_{s_k} = \sum_{i=1}^{n} a_i^{(s_k)} e_i$ such that $a_i^{(s_k)} \to b_i$ for each $1 \leqslant i \leqslant n$. By Lemma §2, $v_{s_k} = \sum_{i=1}^{n} a_i^{(s_k)} e_i$ converges in $(X, \|\cdot\|)$ to $v = \sum_{i=1}^{n} b_i e_i$, and thus $\|v\| = \lim \|v_{s_k}\| = 0$ by Theorem 2.8(4) and since $\|v_m\| \to 0$. This implies $v = 0_X$; since $(e_1, \dots, e_n)$ is a basis of X, we get $b_i = 0$ for each $1 \leqslant i \leqslant n$. Using Remark before this theorem, it follows $\|v_{s_k} - v\|_1 \to 0$, hence, applying AOL Theorem 2.8(4) again, we get $1 = \|v_{s_k}\|_1 = \sum_{i=1}^{n} |a_i^{(s_k)}| \to \sum_{i=1}^{n} |b_i|$, a contradiction since $\sum_{i=1}^{n} |b_i| = 0$. Thus, a desired α exists. □

Examples 2.15 (Sequence of functions converges in one norm but not in the other)**.**

QUICK POLLS: 2. NORMED SPACES See PowerPoint presentation 2, questions 5,6.
Let $f_n(t) = t^n \in C[0,1]$. Then decide whether the sequence f_n converges in

$(C[0,1], \|\cdot\|_1)$? In $(C[0,1], \|\cdot\|_\infty)$?

Corollary 2.16. *The norms $\|\cdot\|_1$ and $\|\cdot\|_\infty$ on $C[a,b]$ are not equivalent.*

Proof. Assume, without loss of generality, that $a = 0$ and $b = 1$, and consider the sequence of functions $f_n(t) = t^n$ as in the Examples 2.15. Let $X = C[0,1]$, $v_n = f_n$, $\|\cdot\| = \|\cdot\|_1$ and $|||\cdot||| = \|\cdot\|_\infty$, $v = 0$. We have just checked that $v_n \to v$ in $\|\cdot\|_1$, and $v_n \not\to v$ in $\|\cdot\|_\infty$.

Theorem 2.10 implies that
if $\|\cdot\|_1 \sim \|\cdot\|_\infty$ then $v_n \to v$ in $\|\cdot\|_1$ implies $v_n \to v$ in $\|\cdot\|_\infty$.

Hence these two norms are not equivalent. □

Lemma 2.17. *The following spaces are infinite-dimensional: ℓ^p for every $1 \leqslant p \leqslant \infty$, c_0, c_{00} and $C[a,b]$ for every $a < b$.*

Proof. We first show that c_{00} is infinite-dimensional. Indeed, Examples 2.7(2) provide us with a sequence which converges to the zero vector in $(c_{00}, \|\cdot\|_\infty)$ but does not converge to the zero vector in $(c_{00}, \|\cdot\|_1)$. By Theorem 2.10 this implies that these two norms on c_{00} are not equivalent, and so by Theorem 2.14 c_{00} is infinite-dimensional.

Since $c_{00} \subseteq \ell^p$ and $c_{00} \subseteq c_0$, this implies that ℓ^p and c_0 are infinite-dimensional too.

By Corollary 2.16, we know that there are two non-equivalent norms on $C[a,b]$. We hence conclude by Theorem 2.14 that $C[a,b]$ is infinite-dimensional. □

Open and closed sets

The following notion is of special importance in the normed spaces.

Definition (Open balls and open and closed sets)

Let $(X, \|\cdot\|)$ be a normed vector space. If $x \in X$ and $r \in (0, \infty)$ then we define
$B(x,r) := \{y \in X : \|y - x\| < r\}$ and call it the *open ball* of radius r and centre x.

Let $U, F \subseteq X$. U is said to be *open* if for each $y \in U$ there exists $d > 0$ (it may depend on x) such that $B(y, d) \subseteq U$.

F is said to be *closed* if $U := X \setminus F$ is open.

Lemma 2.18. *In any normed space, open balls are open sets.*

Proof. Let $x \in X$, $r > 0$, $y \in B(x, r)$. This means that $\|y - x\| < r$. Let $d = r - \|y - x\|$, then $d > 0$. Moreover, if $z \in B(y, d)$ then $\|z - x\| = \|(z - y) + (y - x)\| \leqslant \|z - y\| + \|y - x\| < d + \|y - x\| = r$, hence $z \in B(x, r)$. We proved that for every $y \in B(x, r)$ there exists $d > 0$ such that $B(y, d) \subseteq B(x, r)$. □

Examples (Closed subsets of normed spaces)

1. Let $(X, \|\cdot\|) = (\mathbb{R}^n, \|\cdot\|_2)$, $p_1, \dots, p_n$ be positive real numbers. Then the set

$$W = \{x = (x_1, \dots, x_n) \in X : \text{ there is an index } i \text{ such that } |x_i| \geqslant p_i\} \quad \text{is closed in } X.$$

Proof. To show W is closed we prove that its complement, $U = \mathbb{R}^n \setminus W$, is open. We need to show that $\forall\, x = (x_1, \dots, x_n) \in U$, $\exists\, r > 0$ s.t. $B(x, r) \subseteq U$.

Fix $x = (x_1, \dots, x_n) \in U$. Then $-p_i < x_i < p_i$ for each $i = 1, \dots, n$. Define

$$r = \frac{1}{2} \min\{x_i + p_i, p_i - x_i : i = 1, \dots, n\}.$$

Then $r > 0$ and $-p_i + r < x_i < p_i - r$ for every $i = 1, \dots, n$. We now show that $B(x, r) \subseteq U$. Let $y = (y_1, \dots, y_n) \in B(x, r)$ be arbitrary. We need to prove $y \in U$.

Indeed, $|x_k - y_k| \leqslant \|x - y\|_2 < r$ for each $1 \leqslant k \leqslant n$ (by Lemma 2.12), so

$$-p_k < x_k - r < y_k < x_k + r < p_k.$$

Hence $-p_k < y_k < p_k$ for each $k = 1, \dots, n$, so $y \in U$. □

Subsets of infinite-dimensional spaces need a lot of space to be open!

2. Let $(X, \|\cdot\|) = (\ell^2, \|\cdot\|_2)$.

Is the set $V = \{x = (x_1, x_2, \dots, x_k, \dots) :$ there is an index $k \geqslant 1$ such that $|x_k| \geqslant \frac{1}{k}\}$ closed in $(\ell^2, \|\cdot\|_2)$? (Think about $U = \ell^2 \setminus V$ whether you can fit a ball inside U around $(0, 0, 0, \dots)$.)

QUICK POLLS: 2. NORMED SPACES See PowerPoint presentation 2, question 7.

The following lemma will be very important for us as it tells us how to verify that a particular subset of a normed space is closed.

Lemma 2.19. *Let $(X, \|\cdot\|)$ be a normed vector space and $F \subseteq X$. Then F is closed in X if and only if for any convergent sequence $(v_n)_{n \geqslant 1} \subseteq X$ with limit $v \in X$ we have that if $v_n \in F$ for all $n \geqslant 1$, then $v \in F$ too (F satisfies the **Sequence property**).*

Proof. 1. Assume F is closed and $v_n \in F$ converges to $v \in X$. Let $U = X \setminus F$, by definition of a closed set we have that U is open. If $v \notin F$, then $v \in U$. As U is open, there exists $r > 0$ such that $B(x, r) \subseteq U$. Let N be such that $\|v_n - v\| < r$ for all $n \geqslant N$. Then for all such indices n, we have $v_n \in B(v, r) \subseteq U$, i.e. $v_n \notin F$, a contradiction. This proves $v \in F$.

2. Now assume that the sequence property is satisfied for F. Denote $U = X \setminus F$ and let $v \in U$ be arbitrary (i.e. $v \notin F$). We need to show that there exists $r > 0$ such that $B(v, r) \subseteq U$. If there is no such r, then for every positive number $r_n = 1/n$ there exists $y_n \in B(v, 1/n)$ such that $y_n \notin U$, i.e. $y_n \in F$.

Notice that $y_n \in B(v, 1/n)$ is equivalent to $\|y_n - v\| < 1/n$ for all $n \geqslant 1$. This implies $y_n \to v$, i.e. (y_n) is a convergent sequence with limit v. However $y_n \in F$ for all $n \geqslant 1$, hence by the sequence property $v \in F$, a contradiction with the choice $v \in U = X \setminus F$. □

Remark. We can show that $V \subseteq \ell^2$ is not closed in $(\ell^2, \|\cdot\|_2)$ using the Sequence property. Consider a sequence of vectors $v_n = \frac{1}{n} e_n$, then $v_n \in V$ for each n. However, $\|v_n\|_2 = \frac{1}{n} \to 0$, so $v_n \to \mathbf{0} = (0, 0, 0, \dots)$. Note that $\mathbf{0} \notin V$. By Lemma 2.19, V is not closed.

§3 Banach spaces

Definition (Banach space)

Let $(X, \|\cdot\|)$ be a normed vector space. We say that $(X, \|\cdot\|)$ is a *Banach space* if every Cauchy sequence in X converges.

Remark. When $X = \mathbb{R}$ (with the corresponding absolute value function as the norm), Cauchy's convergence principle states that the converse of Proposition 2.9 is true. This relies on the completeness axiom for the real numbers. However, for general normed vector spaces the converse of Proposition 2.9 is not necessarily true.

Examples 3.1 (Banach spaces)**.** We will work towards the following examples.

1. $X = \mathbb{R}$ with $\|\cdot\| = |\cdot|$ is a Banach space – this is completeness axiom in $\mathbb{R}$.
2. Any finite-dimensional space with any norm is Banach. See Theorem 3.2.
3. For $p \in [1, \infty]$, the space $(\ell^p, \|\cdot\|_p)$ is a Banach space. See Theorem 3.3, proof will not be required for the exam.
4. For $a, b \in \mathbb{R}$, the space $(C[a,b], \|\cdot\|_\infty)$ is a Banach space. However, $(C[a,b], \|\cdot\|_1)$ is not a Banach space. See Theorem 3.6 and Theorem 3.8.
5. As we showed in the Remark after Corollary 2.4, the space $(c_{00}, \|\cdot\|_p)$ is a normed space for all $1 \leqslant p \leqslant \infty$. This space is not a Banach space for any $1 \leqslant p \leqslant \infty$. See Corollary 3.5.

Theorem 3.2. *Any finite-dimensional normed space is a Banach space.*

Proof. Let $(X, \|\cdot\|)$ be a finite-dimensional normed space and $\{e_1, \ldots, e_k\}$ be its basis. Then by Theorem 2.14 we have $\|\cdot\| \sim \|\cdot\|_1$, where $\|\sum_{i=1}^k a_i e_i\|_1 = \sum_{i=1}^k |a_i|$ is defined as in Examples 2.2(2).

Let $(v_n)_{n\geqslant 1}$ be a Cauchy sequence in $(X, \|\cdot\|)$; let $\beta > 0$ be such that $\|x\|_1 \leqslant \beta\|x\|$ for all $x \in X$. Then $(v_n)_{n\geqslant 1}$ is a Cauchy sequence in $(X, \|\cdot\|_1)$ as for any $\varepsilon > 0$ there is $N_{\varepsilon/\beta}$ such that $\|v_n - v_m\|_1 \leqslant \beta(\varepsilon/\beta) = \varepsilon$, for all $m, n \geqslant N_{\varepsilon/\beta}$.

Since $|a_j| \leqslant \|\sum_{i=1}^k a_i e_i\|_1$ for all $1 \leqslant j \leqslant k$, the sequence of jth coordinates of $(v_n)_{n\geqslant 1}$ is Cauchy, for each fixed j. Hence the coordinate sequences all converge. By Remark preceding Theorem 2.14 this implies $(v_n)_{n\geqslant 1}$ converges in $(X, \|\cdot\|)$, since X is finite-dimensional. □

Remark. Theorem 3.2 holds for vector spaces over $\mathbb{R}$ and over $\mathbb{C}$.

Theorem 3.3. *For all* $1 \leqslant p \leqslant +\infty$, $(\ell^p, \|\cdot\|_p)$ *is a Banach space.*

Proof. See Appendix A in §8 for more details. This proof is not required for exam. □

Proposition 3.4. *Let* $(X, \|\cdot\|)$ *be a normed space and let* Y *be a linear subspace of* X *such that* $(Y, \|\cdot\|)$ *is Banach. Then* Y *is a closed subset of* X.

Proof. It is enough to verify that the sequence property from Lemma 2.19 is satisfied for Y.

Assume $(y_n)_{n\geqslant 1} \subseteq Y$ is a convergent sequence in $(X, \|\cdot\|)$ with limit $y \in X$, i.e. $\|y_n - y\| \to 0$. We need to show that $y \in Y$. By Proposition 2.9 we have that (y_n) is Cauchy in $(X, \|\cdot\|)$. However the Cauchy property depends only on values of $\|y_n - y_m\|$, and as $\|y_n - y_m\|_Y = \|y_n - y_m\|$, we get that (y_n) is a Cauchy sequence in $(Y, \|\cdot\|)$ too. As $(Y, \|\cdot\|)$ is a Banach space, any Cauchy sequence converges there, hence there is $v \in Y \subseteq X$ such that $\|y_n - v\| \to 0$. We now have $y = v$ from the uniqueness of limits, Theorem 2.8(1), so $y = v \in Y$. □

Remark. In fact, any closed linear subspace of a Banach space is always a Banach space with the norm from the ambient space, see Q15 of Problem Sheet 2.

Corollary 3.5. *The space* $(c_{00}, \|\cdot\|_p)$ *is not a Banach space, for every* $1 \leqslant p \leqslant \infty$.

Proof. Let $X = \ell^p$, $\|\cdot\| = \|\cdot\|_p$ and $Y = c_{00}$.

Define a sequence $(v_n)_{n\in\mathbb{N}} \subseteq c_{00}$ by

$$v_n = (\frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2^n}, 0, 0, 0, \dots). \tag{3.1}$$

It is immediate that $v_n \in c_{00}$ for every $n \geqslant 1$ as all its coordinates starting after the nth are equal to zero. Furthermore, define $v = (\frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2^n}, \frac{1}{2^{n+1}}, \dots) \in \ell^p$. It is easy to check that $v \in \ell^p$ for each $p \geqslant 1$ a real number and for $p = \infty$. To see that $\|v - v_n\|_p \to 0$, we can estimate, for $p \geqslant 1$ a real number,

$$\|v - v_n\|_p = \Big(\sum_{i=n+1}^{\infty} \Big(\frac{1}{2^k}\Big)^p\Big)^{1/p} = \Big(\sum_{i=n+1}^{\infty} \frac{1}{2^{kp}}\Big)^{1/p} \leqslant \Big(\sum_{i=n+1}^{\infty} \frac{1}{2^k}\Big)^{1/p} = \frac{1}{2^{n/p}} \to 0,$$

and $\|v - v_n\|_\infty = \frac{1}{2^{n+1}} \to 0$.

Clearly $v \notin Y$, hence by Lemma 2.19 Y is not closed in $(X, \|\cdot\|)$. Hence, by Proposition 3.4 we conclude that $(Y, \|\cdot\|) = (c_{00}, \|\cdot\|_1)$ is not a Banach space. □

QUICK POLLS: 3. BANACH SPACES See PowerPoint presentation 3, question 1. Is the sequence $(v_n) \subseteq c_{00}$ from (3.1) a Cauchy sequence?

Non-convergent Cauchy sequences may exist in non-Banach spaces

We now turn our attention to the space of continuous functions with two different norms. First, we consider the case of sup-norm.

Theorem 3.6. *The space* $(C[a, b], \|\cdot\|_\infty)$ *is a Banach space.*

Proof. Let $(f_n) \subseteq (C[a, b], \|\cdot\|_\infty)$ be a Cauchy sequence. This means that for every $\varepsilon > 0$ there is an $N_\varepsilon \in \mathbb{N}$ such that for all $n, m \geqslant N_\varepsilon$ it holds $\|f_n - f_m\|_\infty < \varepsilon$.

Fix any $t_0 \in [a,b]$. We notice that the sequence of real numbers $(f_n(t_0))_{n\geqslant 1} \subseteq \mathbb{R}$ is Cauchy as for every $n, m \geqslant N_\varepsilon$ it holds $|f_n(t_0) - f_m(t_0)| \leqslant \|f_n - f_m\|_\infty < \varepsilon$.

Therefore, for every t, the sequence of real numbers $(f_n(t))_{n\geqslant 1}$ converges.

Denote its limit as $f(t)$. We thus define a function $f : [a,b] \to \mathbb{R}$. We need to show that
(i) f is continuous at every $t_0 \in [a,b]$ and **(ii)** $\|f_n - f\|_\infty = \sup\limits_{t\in[a,b]} |f_n(t) - f(t)| \underset{n\to\infty}{\to} 0$.

Fix $\varepsilon > 0$. Let $n \geqslant N_\varepsilon$ and $t \in [0,1]$ be fixed too.
Then $\forall m \geqslant N_\varepsilon$, $|f_n(t) - f_m(t)| \leqslant \|f_n - f_m\|_\infty < \varepsilon$. Letting m tend to $+\infty$ we get:

$$(\forall \varepsilon > 0\ \exists N_\varepsilon \geqslant 1 \text{ s.t.}) \quad \text{for every } t \in [a,b],\ \text{for every } n \geqslant N_\varepsilon : \quad |f_n(t) - f(t)| \leqslant \varepsilon. \qquad (3.2)$$

This means that the sequence of continuous functions f_n *uniformly converges* to f (see Real Analysis, Year 2), hence f is continuous, this implies (i).

Also, from (3.2) $\|f_n - f\|_\infty = \sup_{t\in[a,b]} |f_n(t) - f(t)| \leqslant \varepsilon$ for all $n \geqslant N_\varepsilon$, i.e., (ii) is satisfied. □

Example 3.7. *Consider a sequence of continuous functions* $g_n(t) = \begin{cases} t^n, & \text{if } 0 \leqslant t \leqslant 1; \\ 1, & \text{if } 1 \leqslant t \leqslant 2. \end{cases}$
It is clear that $g_n \in C[0,2]$ *for all* $n \geqslant 1$.

Cauchy sequences of continuous functions may not converge.

QUICK POLLS: 3. BANACH SPACES See PowerPoint presentation 3, questions 2,3.

Is $(g_n)_{n\geqslant 1}$ a Cauchy sequence in $(C[0,2], \|\cdot\|_1)$?

Theorem 3.8. $(C[a,b], \|\cdot\|_1)$ *is not a Banach space.*

Proof. Consider a sequence of functions g_n as above: we have a Cauchy sequence which does not converge. See Appendix B in §9 for more details. □

§4 Inner product spaces

Definition (Inner product)

Let X be a vector space over $\mathbb{F}$ where $\mathbb{F} = \mathbb{C}$ or $\mathbb{F} = \mathbb{R}$. An *inner product* on X is a mapping $\langle\cdot,\cdot\rangle : X \times X \to \mathbb{F}$ such that

1) For all $x \in X$, the value of $\langle x, x\rangle$ is a real and non-negative number, and $\langle x, x\rangle = 0$ implies $x = 0$;

2) $\langle \alpha x + \beta y, z\rangle = \alpha\langle x, z\rangle + \beta\langle y, z\rangle$ for all $\alpha, \beta \in \mathbb{F}$ and all $x, y, z \in X$;

3) $\langle y, x\rangle = \overline{\langle x, y\rangle}$ for all $x, y \in X$.

The pair $(X, \langle\cdot,\cdot\rangle)$ is called an *inner product space*.

Remark. On a complex vector space, the values of the inner product are complex, on a real vector space, the values of the inner product are all real.

Hence, if $\mathbb{F} = \mathbb{R}$, then last property of the inner product simply says that $\langle x, y\rangle = \langle y, x\rangle$ for all $x, y \in X$.

Examples 4.1. These are examples of standard inner product:

(1) $\mathbb{R}^n = \{(a_1, \dots, a_n) : a_i \in \mathbb{R}\}$, $\langle (a_1, \dots, a_n), (b_1, \dots, b_n)\rangle := \sum_{i=1}^n a_i b_i$.
$\mathbb{C}^n = \{(z_1, \dots, z_n) : z_i \in \mathbb{C}\}$, $\langle (z_1, \dots, z_n), (w_1, \dots, w_n)\rangle := \sum_{i=1}^n z_i \overline{w}_i$.
It is clear that the finite sum is always well-defined, so this is a function on $X \times X$ with values in $\mathbb{F}$.

(2) $\ell^2 = \{(a_i)_{i\geqslant 1} : a_i \in \mathbb{R} \text{ for all } i \geqslant 1, \sum_{i=1}^\infty |a_i|^2 < \infty\}$, or
$\ell^2 = \{(z_i)_{i\geqslant 1} : z_i \in \mathbb{C} \text{ for all } i \geqslant 1, \sum_{i=1}^\infty |z_i|^2 < \infty\}$.
$\big\langle (z_i), (w_i)\big\rangle := \sum_{i=1}^\infty z_i \overline{w}_i$.
We will see in Theorem 4.2 implies that the series $\sum_{i=1}^\infty z_i \overline{w}_i$ converges whenever $(z_i), (w_i) \in \ell^2$.

(3) $C[a, b] = \{f : [a, b] \to \mathbb{F} \text{ continuous}\}$, where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$.
$\langle f, g\rangle := \int_a^b f(t)\overline{g(t)}dt$.

The function $f(t)\overline{g(t)}$ is continuous, hence Riemann integrable.

Example (Inner product of two functions)

Calculate inner products of functions

QUICK POLLS: 4. INNER PRODUCT SPACES PowerPoint presentation 4, questions $1 - 4$.

Theorem 4.2 (Hölder's inequality)**.** *Let* $p, q \in [1, \infty]$ *be such that* $1/p + 1/q = 1$ *(*p *and* q *are conjugate indices). Suppose* $(a_k)_{k\geqslant 1} \in \ell^p$ *and* $(b_k)_{k\geqslant 1} \in \ell^q$*. Then* $(a_k b_k)_{k\geqslant 1} \in \ell^1$ *and*

$$\|(a_k b_k)_{k\geqslant 1}\|_1 \leqslant \|(a_k)_{k\geqslant 1}\|_p \|(b_k)_{k\geqslant 1}\|_q. \tag{4.1}$$

Remark. 1. If $p = q = 2$, then condition $1/p+1/q = 1$ is satisfied. Assume $(z_i)_{i\geqslant 1}, (w_i)_{i\geqslant 1} \in \ell^2$. Then $(\overline{w}_i)_{i\geqslant 1} \in \ell^2$ too as $|\overline{w}_i|^2 = |w_i|^2$. Theorem 4.2 applied to $(z_i)_{i\geqslant 1}$ and $(\overline{w}_i)_{i\geqslant 1}$ gives that $(z_i\overline{w}_i)_{i\geqslant 1} \in \ell^1$, hence $\sum_{i\geqslant 1} |z_i\overline{w}_i|$ converges. This means that the series $\sum_{i\geqslant 1} z_i\overline{w}_i$ is an absolutely convergent series. This implies that the series $\sum_{i\geqslant 1} z_i\overline{w}_i$ converges, hence the inner product $\langle (z_i)_{i\geqslant 1}, (w_i)_{i\geqslant 1}\rangle$ is well-defined.

2. If $p = \infty$, then we interpret the condition $1/p + 1/q = 1$ as $1/q = 1$, i.e., $q = 1$. Vice versa, if $p = 1$, then $q = \infty$ in Theorem 4.2.

Proof of Theorem 4.2 (Hölder's inequality). We will consider cases when one of the indices is ∞ and when both of the are finite, separately.

Case 1: $p = \infty$ and $q = 1$. Then $(a_k)_{k\geqslant 1} \in \ell^\infty$ and so $|a_n| \leqslant \|(a_k)_{k\geqslant 1}\|_\infty$ for all $n \geqslant 1$, by Lemma 2.12 applied for $p = \infty$. Hence $|a_n b_n| \leqslant \|(a_k)_{k\geqslant 1}\|_\infty |b_n|$ for every $n \geqslant 1$, therefore, since the series $\sum_{n\geqslant 1} |b_n|$ converges, the series $\sum_{n\geqslant 1} |a_n b_n|$ converges too (Comparison Test for series) and

$$\|(a_n b_n)\|_1 = \sum_{n\geqslant 1} |a_n b_n| \leqslant \|(a_k)_{k\geqslant 1}\|_\infty \sum_{n\geqslant 1} |b_n| = \|(a_k)_{k\geqslant 1}\|_\infty \|(b_k)_{k\geqslant 1}\|_1.$$

Case 2: $p, q \in (1, \infty)$. Note first that if, say, $\|(a_k)_{k\geqslant 1}\|_p = 0$ then $a_n = 0$ for all $n \geqslant 1$, hence $a_n b_n = 0$ for all $n \geqslant 1$, thus the Hölder inequality is trivial. Assume therefore that both $A = \|(a_k)_{k\geqslant 1}\|_p \neq 0$ and $B = \|(b_k)_{k\geqslant 1}\|_q \neq 0$.

We claim that for all $s, t \in [0, \infty)$ and $p, q > 1$ such that $1/p + 1/q = 1$ it holds

$$st \leqslant \frac{s^p}{p} + \frac{t^q}{q}. \tag{4.2}$$

Inequality (4.2) is referred to as Young's inequality and is proved in Appendix C §10.

Let

$$c_k = \frac{|a_k|}{A}, \quad d_k = \frac{|b_k|}{B}.$$

Note that in the same way as in the proof of Proposition 2.3, $\sum c_k^p = 1$ and $\sum d_k^q = 1$.

Apply now (4.2) to c_k and d_k: $c_k d_k \leqslant \frac{c_k^p}{p} + \frac{d_k^q}{q}$. Summing over $k \in \mathbb{N}$ the last inequality and using $\sum c_k^p = 1$ and $\sum d_k^q = 1$:

$$\frac{1}{AB} \sum_{n\in\mathbb{N}} |a_k b_k| \leqslant \frac{1}{p} + \frac{1}{q} = 1.$$

We get the Hölder's inequality if we multiply both sides of the latter inequality by AB. □

Proof of Examples 4.1: We have already verified that the expressions for standard inner products always produce a well-defined number. We now verify the three properties of the inner product. We do it for Examples 1 and 2 first, then to 3.

[1] and [2]:
1) Note that $z\bar{z} = |z|^2$ if $z \in \mathbb{C}$, hence for $(z_k)_{k\geqslant 1} \in \ell^2$

$$\langle (z_k)_{k\geqslant 1}, (z_k)_{k\geqslant 1} \rangle = \sum_{n=1}^{\infty} z_n \bar{z}_n = \sum_{n=1}^{\infty} |z_n|^2, \tag{4.3}$$

a non-negative real number, and if $\langle (z_k)_{k\geqslant 1}, (z_k)_{k\geqslant 1} \rangle = 0$, then all $|z_n|^2 = 0$, hence all $z_n = 0$, hence $(z_k)_{k\geqslant 1} = (0, 0, 0, 0, 0, 0, \dots)$ =zero vector-sequence.

2) $\langle \alpha(z_k)_{k\geqslant 1} + \beta(y_k)_{k\geqslant 1}, (w_k)_{k\geqslant 1} \rangle = \langle (\alpha z_k + \beta y_k)_{k\geqslant 1}, (w_k)_{k\geqslant 1} \rangle = \sum_{k=1}^{\infty} (\alpha z_k + \beta y_k)\overline{w}_k = \alpha \sum_{k=1}^{\infty} z_k \overline{w}_k + \beta \sum_{k=1}^{\infty} y_k \overline{w}_k = \alpha \langle (z_k)_{k\geqslant 1}, (w_k)_{k\geqslant 1} \rangle + \beta \langle (z_k)_{k\geqslant 1}, (w_k)_{k\geqslant 1} \rangle$.

3) $\langle (y_k)_{k\geqslant 1}, (z_k)_{k\geqslant 1}\rangle = \sum_{k=1}^{\infty} y_k\overline{z}_k = \sum_{k=1}^{\infty}\overline{\overline{y}_k z_k} = \overline{\sum_{k=1}^{\infty}\overline{y}_k z_k} = \langle (z_k)_{k\geqslant 1}, (y_k)_{k\geqslant 1}\rangle$.

[3]:

$$\langle f, f\rangle = \int_a^b f(t)\overline{f(t)}\,dt = \int_a^b |f(t)|^2\,dt, \tag{4.4}$$

which is a real non-negative number. Moreover, if $\langle f, f\rangle = 0$, then the integral norm of the continuous real-valued non-negative function $F(t) = |f(t)|^2$ is equal to 0, hence $F(t) = 0$ for all $t \in [a, b]$, which implies $f(t) = 0$ for all $t \in [a, b]$.

Verification of two remaining properties of the inner product for the formula given in Example [3] is left as exercise. □

Recall the result in (4.3):

- In ℓ^2, $\langle x, x\rangle = \sum\limits_{n=1}^{\infty} |x_n|^2 = (\|x\|_2)^2$ - the square of the norm of x.

It turns out that this behaviour is general to all inner product spaces.

If X is an inner product space, we claim that the following mapping defines a norm on X:

$$\|\cdot\| : X \to [0, \infty); \quad x \mapsto \langle x, x\rangle^{1/2} \tag{4.5}$$

This norm is called the *norm induced by the inner product* or simply *induced norm*.

Lemma 4.3. *Let $(X, \langle\cdot,\cdot\rangle)$ be an inner product space. Let $\|x\| = \langle x, x\rangle^{1/2}$ for each $x \in X$. Then $\|\cdot\| : X \to \mathbb{R}$ is well-defined and satisfies the first two properties of a norm.*

Proof. As $\langle x, x\rangle \geqslant 0$ is a real number, the square root $\sqrt{\langle x, x\rangle}$ is always well-defined.

(i) $\|x\| = \langle x, x\rangle^{1/2} \geqslant 0$; if $\|x\| = 0$ then $\langle x, x\rangle = 0$ which implies $x = 0$;

(ii) $\|\lambda x\| = (\langle \lambda x, \lambda x\rangle)^{1/2} = \Big(\lambda\langle x, \lambda x\rangle\Big)^{1/2} = \Big(\lambda\overline{\langle \lambda x, x\rangle}\Big)^{1/2} = \Big(\lambda\overline{\lambda}\overline{\langle x, x\rangle}\Big)^{1/2} = \Big(\lambda\overline{\lambda}\cdot\overline{\langle x, x\rangle}\Big)^{1/2} = \Big(\lambda\overline{\lambda}\langle x, x\rangle\Big)^{1/2} = \Big(|\lambda|^2\langle x, x\rangle\Big)^{1/2} = |\lambda|\|x\|$. □

Definition (Orthogonal vectors, collections)

Let $(X, \langle\cdot,\cdot\rangle)$ be an inner product space. We write $x \perp y$ and say x is *orthogonal* to y if $\langle x, y\rangle = 0$. We say $S \subset X$ is an *orthogonal collection* if $x \perp y$ for all distinct $x, y \in S$.

Theorem 4.4 (Pythagoras' theorem)**.** *Let $(X, \langle\cdot,\cdot\rangle)$ be an inner product space.*

(1) If $x \perp y$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

(2) If a finite collection of vectors $\{v_1, \ldots, v_n\}$ is orthogonal, then $\|\sum_{k=1}^{n} v_k\|^2 = \sum_{k=1}^{n} \|v_k\|^2$.

Proof. (1) If $\langle x, y\rangle = 0$ then $\langle y, x\rangle = \overline{\langle x, y\rangle} = 0$, so $\|x+y\|^2 = \langle x+y, x+y\rangle = \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle = \langle x, x\rangle + \langle y, y\rangle = \|x\|^2 + \|y\|^2$.

(2) Follows from (1) by induction. □

Theorem 4.5 (Cauchy–Schwarz inequality)**.** *If $(X, \langle\cdot,\cdot\rangle)$ is an inner product space and $\|\cdot\|$ is the induced norm on X, then $|\langle x, y\rangle| \leqslant \|x\| \cdot \|y\|$ for any $x, y \in X$.*

Proof. First note that if $y = 0$, then both sides are equal to 0.

Assume $y \neq 0$, then $\|y\|^2 = \langle y, y\rangle \neq 0$. Let $z = x - \frac{\langle x,y\rangle}{\langle y,y\rangle} y$. Then $\langle z, y\rangle = \langle x, y\rangle - \frac{\langle x,y\rangle}{\langle y,y\rangle}\langle y, y\rangle = 0$. Apply now Pythagoras Theorem 4.4 to orthogonal vectors z and $\frac{\langle x,y\rangle}{\langle y,y\rangle} y$ to get, using Lemma 4.3(ii):

$$\|x\|^2 = \|z\|^2 + \left\|\frac{\langle x, y\rangle}{\langle y, y\rangle} y\right\|^2 \geqslant \left\|\frac{\langle x, y\rangle}{\langle y, y\rangle} y\right\|^2 = \left|\frac{\langle x, y\rangle}{\langle y, y\rangle}\right|^2 \|y\|^2 = \frac{|\langle x, y\rangle|^2}{\|y\|^2}.$$ □

Theorem 4.6. *If $(X, \langle\cdot,\cdot\rangle)$ is an inner product space and $\|\cdot\|$ is an induced norm, then $(X, \|\cdot\|)$ is a normed vector space.*

Proof. Using Lemma 4.3, we only need to verify the triangle inequality. If $x, y \in X$ then by Cauchy–Schwarz, Theorem 4.5, $\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2\mathrm{Re}\langle x, y\rangle \leqslant \|x\|^2 + \|y\|^2 + 2|\langle x, y\rangle|$ $\leqslant \|x\|^2 + \|y\|^2 + 2\|x\|\|y\| = (\|x\| + \|y\|)^2$.

Taking square roots of the LHS and the RHS, we get $\|x+y\| \leqslant \|x\| + \|y\|$. □

Another corollary of the Cauchy–Schwarz inequality is the continuity of the inner product.

Lemma 4.7. *Let (v_n) and (u_n) be two sequences in the inner product space X converging to v and u respectively. Then the sequence $(\langle v_n, u_n\rangle)$ converges to $\langle v, u\rangle$.*

Proof. Apply Theorem 4.5, Cauchy-Schwarz inequality, to get

$$\begin{aligned} 0 \leqslant |\langle v_n, u_n\rangle - \langle v, u\rangle| &= |\langle v_n - v, u_n\rangle + \langle v, u_n - u\rangle| \\ &\leqslant |\langle v_n - v, u_n\rangle| + |\langle v, u_n - u\rangle| \\ &\leqslant \|v_n - v\|\|u_n\| + \|v\|\|u_n - u\| \to 0 \end{aligned}$$

as $n \to \infty$ since $\|u_n\| \to \|u\|$ by Theorem 2.8. □

We now come to a very important statement which shows how the properties of the inner product influence the properties of the induced norm.

Theorem 4.8. *Let $(X, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F}$ and let $\|\cdot\|$ denote the induced norm. Then*

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2 \tag{4.6}$$

for all $x, y \in X$. We call this the Parallelogram law.

Proof. If $(X, \langle\cdot,\cdot\rangle)$ is an inner product space then one can show (4.6) holds by expanding the left-hand side using the linearity of the inner product:

$\|x+y\|^2 + \|x-y\|^2 = \langle x+y, x+y\rangle + \langle x-y, x-y\rangle$
$= \langle x,x\rangle + \langle x,y\rangle + \langle y,x\rangle + \langle y,y\rangle + \langle x,x\rangle - \langle x,y\rangle - \langle y,x\rangle + \langle y,y\rangle = 2\|x\|^2 + 2\|y\|^2.$ □

Remark. We have seen that the space $(\ell^2, \|\cdot\|_2)$ has inner product which induces its norm. But what about other $(\ell^p, \|\cdot\|_p)$? It turns out that for $p \neq 2$ these spaces cannot be made inner product spaces. We now explain what we mean by this.

Example (Spaces without inner product)

In the space $(\ell^\infty, \|\cdot\|_\infty)$ there does not exist an inner product whose induced norm is $\|\cdot\|_\infty$.

Proof. Consider $x = (1,0,0,\dots)$, $y = (0,1,0,0,\dots)$.

Then $x, y \in \ell^\infty$, $\|x\|_\infty = 1$, $\|y\|_\infty = 1$, $\|x+y\|_\infty = 1$, $\|x-y\|_\infty = 1$. The conclusion of Theorem 4.8 (the parallelogram law) is not satisfied as $1^2 + 1^2 \neq 2(1^2+1^2)$, hence there is no inner product on ℓ^∞ which would induce $\|\cdot\|_\infty$.

In the case $1 \leqslant p < \infty$, $p \neq 2$ (Question 19 of Problem Sheet 2) we may also show that there is no inner product on ℓ^p which would induce $\|\cdot\|_p$. □

In fact, the following theorem is true. We provide its proof in Appendix C §11. Its proof is not examinable.

Theorem 4.9. *Let $(X, \|\cdot\|)$ be a normed vector space such that the Parallelogram law* (4.6) *holds for any $x, y \in X$. Then there exists an inner product $\langle\cdot,\cdot\rangle : X \times X \to \mathbb{F}$ which induces the norm $\|\cdot\|$; i.e. $\langle x,x\rangle^{1/2} = \|x\|$ for each $x \in X$.*

Example (Orthogonal collections)

[1] $X = \mathbb{R}^3$ with standard inner product (dot product).

$S = \{(1,0,0), (0,1,0), (0,0,1)\}$ is an orthogonal collection and every vector has norm 1.

An orthogonal collection of vectors $S \subseteq X$ is called *orthonormal* if $\|x\| = 1$ for any $x \in S$. (Note that this condition is equivalent to $\langle x,x\rangle = 1$ for all $x \in S$.)

[2] $X = \ell^2$ with standard inner product.

Then $S = \{e_n : n \geqslant 1\}$, where $e_n = (\underbrace{0,\dots,0}_{n-1}, 1, 0, 0, \dots)$, is an orthonormal collection.

[3] Let $X = C[0, 2\pi]$ be a space of complex-valued continuous functions with the standard inner product.

Then $S = \left\{\frac{1}{\sqrt{2\pi}} e^{int} \mid n \in \mathbb{Z}\right\}$ is an orthonormal collection.

Let $f \neq g$ be two elements of S. Then there exist $n \neq m$ integers such that $f(t) = \frac{1}{\sqrt{2\pi}} e^{int}$ and $g(t) = \frac{1}{\sqrt{2\pi}} e^{imt}$. Then

$$\langle f, g \rangle = \int_0^{2\pi} f(t)\overline{g(t)}\, dt = \int_0^{2\pi} \frac{1}{\sqrt{2\pi}} e^{int} \overline{\frac{1}{\sqrt{2\pi}} e^{imt}} dt$$
$$= \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)t} = \frac{1}{2\pi} \frac{1}{i(n-m)} e^{i(n-m)t} \Big|_0^{2\pi} = \frac{1}{2\pi} \frac{1}{i(n-m)} (1-1) = 0,$$

so S is orthogonal. Also, for any $f(t) = \frac{1}{\sqrt{2\pi}} e^{int} \in S$ we have

$$\langle f, f \rangle = \int_0^{2\pi} \frac{1}{\sqrt{2\pi}} f(t) \overline{\frac{1}{\sqrt{2\pi}} f(t)} dt = \frac{1}{2\pi} \int_0^{2\pi} e^{int} e^{-int} dt = \frac{1}{2\pi} \int_0^{2\pi} 1 dt = 1.$$

Hence S is orthonormal.

§5 Linear functionals

Definition (Linear functional)

Let X be a vector space over $\mathbb{F}$. A *linear functional*[1] on a vector space X is a linear mapping $f : X \to \mathbb{F}$:

$$f(\alpha x + y) = \alpha f(x) + f(y) \quad \text{for all} \quad x, y \in X \quad \text{and} \quad \alpha \in \mathbb{F}.$$

Here $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

Also, the collection of all linear functionals on a given vector space X is a vector space itself: the operations of addition $f+g$ and scalar multiplication cf, where f, g are linear functionals and $c \in \mathbb{F}$, are well-defined and satisfy all required axioms.

The zero element of this vector space of linear functionals is the functional $\mathbf{0}$ defined as $\mathbf{0}(x) = 0$ for all $x \in X$.

Remark. 1) Note that the property of $f : X \to \mathbb{F}$ being a linear functional does not depend on any additional structure we may have on X (such as a norm or an inner product) - we only use that X is a vector space.

2) If $w = 0_X \in X$ is the zero vector, then $f(w) = 0_{\mathbb{F}}$. Indeed, $w = w + w$, so $f(w) = f(w+w) = f(w) + f(w) = 2f(w)$, hence $f(w) = 0_{\mathbb{F}}$.

Definition (Bounded linear functional)

A linear functional f on a normed vector space $(X, \|\cdot\|)$ is called a *bounded linear functional* if there exists $C \geqslant 0$ such that $|f(x)| \leqslant C\|x\|$ for each $x \in X$.

[1] A very similar construction is that of *linear operators*. These are linear mappings from a vector space to another vector space. See §7.

Definition (Dual space)

The collection of all bounded linear functionals on X is denoted as X^* or $(X, \|\cdot\|)^*$ and is called the space *dual* to $(X, \|\cdot\|)$.

Note that we need to always keep in mind with respect to which norm $\|\cdot\|$ on X we consider bounded linear functionals. Even though the indication of which norm $\|\cdot\|$ is considered on X is suppressed, the specific norm plays a very important part in defining X^*.

Lemma 5.1. *Let $(X, \|\cdot\|)$ be a normed space. The collection of all bounded linear functionals on X forms a vector space.*

Proof. Since the collection of bounded linear functionals is a subset of the vector space of all linear functionals, we only need to verify two properties:

• Zero functional $f(x) = 0$ for all $x \in X$ is a bounded linear functional: let $C = 0$, then $|f(x)| = 0 \leqslant 1\|x\|$ for all $x \in X$.

• If f_1, f_2 are bounded linear functionals, then $af_1 + f_2$ is a linear functional too. If C_1, C_2 are corresponding constants, then $|(af_1 + f_2)(x)| = |af_1(x) + f_2(x)| \leqslant (|a|C_1 + C_2)\|x\|$, so $af_1 + f_2$ is a bounded linear functional. □

QUICK POLLS: 5. LINEAR FUNCTIONAL See Power Point presentation 5, questions $1-5$.

For a bounded linear functional, it is important which norm is considered on X.

Definition (Norm of a functional)

We already know that X^* is a vector space. We now explain how to define a norm on X^*.

Let $(X, \|\cdot\|)$ be a normed vector space and $f \in X^*$ (a bounded linear functional). Its *norm* is defined as

$$\|f\|_{X^*} = \inf\Big\{C \in [0,\infty) : |f(x)| \leqslant C\|x\| \text{ for all } x \in X\Big\}.$$

Remark. 1. Note first that the set of constants $C \geqslant 0$ such that $|f(x)| \leqslant C\|x\|$ for all $x \in X$ is not empty, this is by the definition of a bounded linear functional. This set is also bounded from below by 0, hence the infimum exists and is a finite real number,

$$\|f\|_{X^*} < \infty \text{ for any } f \in X^*.$$

2. This is indeed a norm on the vector space X^*; see Question 24 of Problem Sheet 2.

Lemma 5.2. *Let $(X, \|\cdot\|)$ be a normed space and $f \in X^*$. Then $|f(x)| \leqslant \|f\|_{X^*}\|x\|$ for all $x \in X$.*

Proof. Let $C_n \searrow \|f\|_{X^*}$ be such that for $|f(x)| \leqslant C_n\|x\|$ for all $x \in X$, for each $n \geqslant 1$. Then we conclude for each fixed $x \in X$ that $|f(x)| \leqslant \|f\|_{X^*}\|x\|$. □

Proposition 5.3. *Let $(X, \|\cdot\|_X)$ be a normed vector space and X^* be the space of bounded linear functionals. Then for any $f \in X^*$*

$$\|f\|_{X^*} = \sup\{|f(x)| : \|x\|_X \leqslant 1\}$$

Proof. Denote $B = \{x : \|x\|_X \leqslant 1\}$, and $S = \sup\{|f(x)| : x \in B\}$. For every $x \in B$, we have $|f(x)| \leqslant \|f\|_{X^*}\|x\|_X \leqslant \|f\|_{X^*}$ by Lemma 5.2, therefore $S \leqslant \|f\|_{X^*}$. To complete the proof it is enough to verify that $S \geqslant \|f\|_{X^*}$.

Fix any $x \in X\backslash\{\mathbf{0}_X\}$. The vector $y = x/\|x\|$ has norm equal to 1, so $y \in B$, therefore, $|f(y)| \leqslant S$. This means $|f(x)| \leqslant S\|x\|$ for every $x \neq \mathbf{0}_X$. This inequality is clearly satisfied for $x = \mathbf{0}_X$ too. Therefore, S is one of the constants C which works for all $x \in X$ for the inequality $|f(x)| \leqslant C\|x\|$. The norm $\|f\|_{x^*}$, being the infimum of all such C, is less than or equal to S.

□

We show an example of calculating a norm of an element of the dual space.

Calculate a norm of a particular bounded linear functional

QUICK POLLS: 5. LINEAR FUNCTIONAL See Power Point presentation 5, questions 6-7.

We finish this chapter with the following important theorem.

Theorem 5.4. *Let f be a linear functional on the normed vector space X. Then the following are equivalent.*

1. *f is a bounded linear functional;*
2. *f is continuous;*
3. *$\ker f = \{v \in X : f(v) = 0\}$ is a closed subset of X;*
4. *f is continuous at the origin, point $x = 0$.*

Proof. We prove $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$. The implications involving (4) are proved in the Question 23 of Problem Sheet 2.

$(1) \Rightarrow (2)$: This follows from the linearity of f:

As f is a bounded linear functional, there exists $C \geqslant 0$ s.t. $|f(x)| \leqslant C\|x\|$ for all $x \in X$.

Let $v_n, v \in X$ be such that $v_n \to v$, i.e. $\|v_n - v\| \to 0$. Then $0 \leqslant |f(v_n) - f(v)| = |f(v_n - v)| \leqslant C\|v_n - v\| \to 0$, so $f(v_n) \to f(v)$.

$(2) \Rightarrow (3)$: We prove that $\ker f$ is closed using Lemma 2.19. Assume $(v_n) \subseteq \ker f$ is a convergent sequence in $(X, \|\cdot\|)$, and v is its limit. Then $v_n \to v$ and $f(v_n) = 0$ for all $n \geqslant 1$. As f is continuous, we get $f(v) = \lim f(v_n) = \lim 0 = 0$, hence $v \in \ker f$. Thus $\ker(f)$ satisfies the sequence property, so is closed.

(3) ⇒ (1): Assume (for a contradiction) that f is not a bounded linear functional, then we get (from the definition) that for every $C > 0$ there exists $x_C \in X$ such that

$$|f(x_C)| > C\|x_C\|. \tag{5.1}$$

Take in particular $C = n \in \mathbb{N}$. We get a sequence $(x_n) \subseteq X$ such that $|f(x_n)| > n\|x_n\|$ for each $n \geqslant 1$. In particular $f(x_n) \neq 0$ for each n. Define $y_n = x_n/f(x_n)$. Then $\|y_n\| = \frac{\|x_n\|}{|f(x_n)|} < 1/n$, so that $y_n \to 0$, and $f(y_n) = 1$ for each n.

Now consider the sequence of vectors $v_n = y_1 - y_n$. Each of the vectors v_n belongs to $\ker f$ as $f(v_n) = f(y_1 - y_n) = f(y_1) - f(y_n) = 1 - 1 = 0$. We can also say that $v_n = y_1 - y_n \to y_1 - 0 = y_1$. However $y_1 \notin \ker f$ as $f(y_1) = 1 \neq 0$. Hence $\ker f$ does not satisfy the sequence property, so, by Lemma 2.19, is not closed. This is a contradiction.

□

§6 Hilbert spaces

Definition (Hilbert space)

An inner product space $(X, \langle\cdot,\cdot\rangle)$ that is a Banach space with respect to the norm induced by the inner product is called a *Hilbert space*.

Example (Hilbert and non-Hilbert spaces)

[1] The finite dimensional spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ have inner product which defines an induced norm (for example, the standard inner product induces $\|\cdot\|_2$). A finite-dimensional space with *any* norm is a Banach space (Theorem 3.2), hence a finite dimensional space with *any* inner product is a Hilbert space.

[2] The infinite-dimensional space ℓ^2 is a Hilbert spaces as the standard inner product induces $\|\cdot\|_2$, and $(\ell^2, \|\cdot\|_2)$ is a Banach space, see Theorem 3.3.

[3] The inner product space $C[a,b]$ with the product $\langle f, g\rangle = \int_a^b f(t)\overline{g(t)}dt$ is not a Hilbert space. This follows from the fact that $(C[a,b], \|\cdot\|_2)$, where

$$\|f\|_2 = \left(\int_a^b |f(t)|^2 dt\right)^{1/2}, \quad f \in C[a,b]$$

is not a Banach space. The proof of this is similar to that of $(C[a,b], \|\cdot\|_1)$ is not a Banach space.

See Appendix A §9 for more details.

Distances and minimising vectors.

We address the question of whether the distance from a vector v of a Hilbert space H to a given subset M,

$$\operatorname{dist}(v, M) := \inf\{\|v - y\| : y \in M\},$$

is attained or not. In other words, the question we want to answer: does there exist a vector $u \in M$ such that $\|v - u\| = \operatorname{dist}(v, M)$? We will call such vector u a *minimising* vector.

Discussion: Existing of minimising vector

Is there a minimising vector in c_{00}?

QUICK POLLS: 6. HILBERT SPACES See Power Point presentation 6, questions 1-2.

Theorem 6.1. *Let $M \neq \emptyset$ be a closed convex subset of the Hilbert space H.*

Then for every $x \in H$ there exists a unique $y \in M$ such that dist$(x, M) = \|x - y\|$.

Discussion: What is wrong with the example discussed before Theorem 6.1?

Why Thm 6.1 does not work for c_{00}?

QUICK POLLS: 6. HILBERT SPACES See Power Point presentation 6, question 3.

We are now ready to prove this theorem.

Proof of Theorem 6.1. Let $d := \operatorname{dist}(x, M)$ then by the definition of infimum there exists a sequence $(v_n) \in M$ such that

$$d_n := \|x - v_n\| \to d. \tag{6.1}$$

We claim that (v_n) is a Cauchy sequence. Indeed, due to convexity of M, we have that the middle of the straight line interval connecting any two members of this sequence is also in M: $(v_n + v_m)/2 \in M$ for any $n, m \geqslant 1$, hence

$$\|(v_n + v_m)/2 - x\| \geqslant d \text{ for all } n, m \geqslant 1.$$

Thus using the Parallelogram Law (4.6) Theorem 4.8:

$$\|u_1 + u_2\|^2 + \|u_1 - u_2\|^2 = 2\|u_1\|^2 + 2\|u_2\|^2$$

for $u_1 = v_n - x$, $u_2 = v_m - x$ we get

$$\begin{aligned} 0 \leqslant \|v_n - v_m\|^2 &= -\|v_n + v_m - 2v\|^2 + 2(\|v_n - v\|^2 + \|v_m - v\|^2) \\ &\leqslant -(2d)^2 + 2(d_n^2 + d_m^2) \to 0 \end{aligned}$$

as $m, n \to \infty$ because of (6.1). We therefore conclude that (v_n) is a Cauchy sequence. Since H is a Banach space, (v_n) converges to some point $y \in H$. As M is closed and $v_n \in M$ for all n we get $y \in M$ by Lemma 2.19.

Moreover, by Theorem 2.8 (4), $\|x - y\| = \lim \|x - v_n\| = \lim d_n = d$. Thus, we found $y \in M$ with $\|x - y\| = d = \operatorname{dist}(x, M)$.

For uniqueness, suppose that $y' \in M$ is such that $d = \|x - y'\|$ as well. Then using convexity of M we have $(y+y')/2 \in M$, hence $d \leqslant \left\| \frac{y+y'}{2} - x \right\|$. By the Parallelogram Law (Theorem 4.8) for $y - x$, $y' - x$ we get

$$\|y - y'\|^2 = -\|y + y' - 2x\|^2 + 2(\|y - x\|^2 + \|y' - x\|^2) \leqslant -(2d)^2 + 2(d^2 + d^2) = 0$$

and so $\|y - y'\| = 0 \implies y = y'$. □

Theorem 6.2. *Let L be a linear subspace of the Hilbert space H, $v \in H$ and $u \in L$.*

Then $\operatorname{dist}(v, L) = \|v - u\|$ (in other words, u is the minimising vector for v) if and only if $(v - u)$ is orthogonal to L, i.e., $(v - u) \perp z$ for every $z \in L$.

Proof. First suppose that $(v - u)$ is orthogonal to L. Then by Pythagoras' theorem (Theorem 4.4) for every $z \in L$ we have

$$\|v - z\|^2 = \|(v - u) + (u - z)\|^2 = \|v - u\|^2 + \|u - z\|^2 \geqslant \|v - u\|^2,$$

hence $\|v-z\| \geqslant \|v-u\|$ as required. Here we used that $u-z \in L$ implies $(v-u) \perp (u-z)$.

Conversely: suppose that $\operatorname{dist}(v, L) = \|v-u\|$. Let $y \in L$ be an arbitrary vector with $\|y\| = 1$; set $\alpha := \langle v-u, y\rangle$. We need to show that $\alpha = 0$.

Note that

$$\|v-(u+\alpha y)\|^2 = \|v-u-\alpha y\|^2 = \Big\langle (v-u)-\alpha y, (v-u)-\alpha y \Big\rangle = \|v-u\|^2 - \alpha\langle y, v-u\rangle - \overline{\alpha}\langle v-u, y\rangle + |\alpha|^2 = \|v-u\|^2 - |\alpha|^2$$

as $\overline{\alpha}\langle v-u, y\rangle = \overline{\alpha}\alpha = |\alpha|^2$ and $\alpha\langle y, v-u\rangle = \alpha\overline{\alpha} = |\alpha|^2$. Since $(u+\alpha y) \in L$ we note $\|v-(u+\alpha y)\| \geqslant \|v-u\|$. This implies $\|v-u\|^2 \leqslant \|v-u\|^2 - |\alpha|^2$, hence $\alpha = 0$.

Then for any[2] $z \in L$ we can write $z = \|z\|y$ with $\|y\| = 1$, therefore $\langle v-u, z\rangle = \|z\|\langle v-u, y\rangle = \|z\| \cdot 0 = 0$. We thus have $\langle v-u, z\rangle = 0$ for all $z \in L$, hence $v-u \perp L$. □

These two theorems allow us to give the following definition.

Definition (Orthogonal projection)

Given a closed linear subspace L of the Hilbert space H and $x \in H$, the minimising vector y arising from Theorem 6.1 and satisfying the conclusions of Theorem 6.2 is said to be the *orthogonal projection* of x on L. This defines a mapping $P_L : H \to L$ by $P_L(x) := y$. It satisfies the condition that $\boxed{(x - P_L(x)) = (x-y) \perp L}$ for every $x \in H$ (orthogonality condition).

Remark. If $x \in L$, then the minimising vector is x itself: $P_L(x) = x$.

QUICK POLLS: 6. HILBERT SPACES See Power Point presentation 6, question 4.

Example 6.3 (Orthogonal projection on a finite-dimensional space).

Suppose L is a linear subspace of the Hilbert space H, and its basis $\{e_1, \dots, e_n\}$ is an orthonormal collection, i.e. these vectors are pairwise orthogonal and norm of each vector is 1. Note that L is finite-dimensional hence Banach (Theorem 3.2). Hence by Proposition 3.4, we have that L is a closed subspace of H. Thus the orthogonal projection $P_L : H \to L$ from H onto a closed linear subspace L is well-defined.

If $x \in H$ then $P_L(x) \in L$, so that $P_L(x) = \sum_{k=1}^n \alpha_k e_k$ for some scalars α_k. By Theorem 6.2 we have $x - P_L(x) \perp L \iff x - P_L(x) \perp e_k \ \forall k = 1, \dots, n \iff \langle x - P_L(x), e_k\rangle = 0 \ \forall k = 1, \dots, n$ $\iff$

$$\langle x, e_k\rangle = \langle P_L(x), e_k\rangle = \Big\langle \sum_{s=1}^n \alpha_s e_s, e_k \Big\rangle = \sum_{s=1}^n \alpha_s \langle e_s, e_k\rangle = \alpha_k \langle e_k, e_k\rangle = \alpha_k.$$

Here we used $\langle e_s, e_k\rangle = 0$, and $\langle e_k, e_k\rangle = 1$ for all $1 \leqslant s \neq k \leqslant n$.

Finally, we get $\boxed{P_L(x) = \sum_{k=1}^n \langle x, e_k\rangle e_k.}$

[2] If $z \neq 0$, then $y = z/\|z\|$, otherwise $\|z\| = 0$ so we can take any y with $\|y\| = 1$.

QUICK POLLS: 6. HILBERT SPACES See Power Point presentation 6, question 5.

The formula for P_L will be used in the next part of this chapter.

Definition (Orthogonal complement)

Let $(X, \langle\cdot,\cdot\rangle)$ be an inner product space and $M \subseteq X$. We call

$$M^\perp = \{y \in X : y \perp M\}$$

the *orthogonal complement* of M in X.

We introduce the notion of a closure of a set. We will only need/use this notion for linear subspaces but note that it works in more general settings too.

Definition (Closure and dense)

Let $(X, \|\cdot\|)$ be a normed space and $S \subseteq X$. The closure of S is the set

$$\overline{S} = \Big\{\lim v_n : (v_n)_{n\geqslant 1} \subseteq S \text{ is a convergent sequence in } X\Big\}.$$

S is called *dense* in $(X, \|\cdot\|)$ if $\overline{S} = X$.

Note that if L is a linear subspace of X, then $\bar{L}$ is a closed linear subspace, by Sequence Property.

Theorem 6.4. *Let M be a subset of the Hilbert space H. Then the linear span $\langle M\rangle$ is dense in H if and only if $M^\perp = \{0\}$.*

Proof. Assume first that the *linear span $\langle M\rangle$ is dense in H* and $x \in M^\perp$. By definition of *dense*, there exists $(v_n)_{n\geqslant 1} \subseteq \langle M\rangle$ such that $v_n \to x$. Since $x \perp M$, we have $x \perp v_n$ for each $n \geqslant 1$ too. Hence using Lemma 4.7, $\langle x, x\rangle = \lim\langle x, v_n\rangle = \lim 0 = 0$; this implies $x = 0$.

Since $M^\perp$ is a linear subspace by Question 16 from PS2, we conclude $0 \in \langle M\rangle$, therefore $M^\perp = \{0\}$.

Assume now $M^\perp = \{0\}$. Let $L = \overline{\langle M\rangle}$; assume $x \in H$ is any. Since L is a closed linear subspace, consider $P_L : H \to L$, the orthogonal projection and $y = P_L(x) \in L$. Since $(x - y) = (x - P_L(x)) \perp L$ by definition of orthogonal projection, we have $(x - y) \in L^\perp = \{0\}$, implying $x - y = 0$, thus $x = y \in L$. Hence $H \subseteq L$, so $L = H$, thus $\langle M\rangle$ is dense in H. □

Remark. If $H = \ell^2$ and $M = \{e_n : n \geqslant 1\}$, then the linear span $\langle M\rangle$, which we proved is equal to c_{00}, is dense in H.

Indeed, by definition of closure, for every $x = (a_n)_{n\geqslant 1} \in \ell^2$ we have $\|x - v_n\|_2 \to 0$ for $v_n = (a_1, \dots, a_n, 0, 0, 0, \dots)$.

We could instead verify the condition from Theorem 6.4: if $x = (a_n)_{n\geqslant 1} \in M^\perp$, then $a_n = \langle x, e_n\rangle = 0$ for each $n \geqslant 1$, so $x = 0$. Thus $M^\perp = \{0\}$.

QUICK POLLS: 6. HILBERT SPACES See Power Point presentation 6, questions 6a,b: Examples on how Theorem 6.4 can be used in order to decide whether the span of a particular subset of ℓ^2 is or isn't dense in $(\ell^2, \|\cdot\|_2)$.

Lemma 6.5. *Let* H *be a Hilbert space and* L *be a closed linear subspace of* H. *Then the orthogonal projection* $P_L : H \to L$ *satisfies* $\|P_L(x)\| \leqslant \|x\|$ *for each* $x \in H$.

Proof. Apply Theorem 6.2 to get $x - P_L(x) \perp P_L(x)$ and therefore by the Pythagoras' theorem (Theorem 4.4), $\|x\|^2 = \|(x - P_L(x)) + P_L(x)\|^2 = \|x - P_L(x)\|^2 + \|P_L(x)\|^2 \geqslant \|P_L(x)\|^2$. □

Theorem 6.6. *Let* H *be a Hilbert space and* $\mathcal{E} = \{v_1, v_2, \dots\}$ *be an infinite orthonormal sequence,* $L = \overline{\langle \mathcal{E} \rangle}$, $P_L : H \to L$ *is the orthogonal projection on* L *and* $x \in H$. *Then*

[1] The sequence of vectors $y_n = \sum_{k=1}^{n} \langle x, v_k \rangle v_k$ *converges to* $P_L(x)$

[2] $\|P_L(x)\|^2 = \sum_{k=1}^{\infty} |\langle x, v_k \rangle|^2$

Proof. For every $N \geqslant 1$, we have that $\mathcal{B}_n = \{v_1, \dots, v_n\}$ is an orthonormal collection in H, let $L_n = \langle \mathcal{B}_n \rangle$ be the closed linear (finite-dimensional) subspace of H and $P_n = P_{L_n}$ be the orthogonal projection[3] onto L_n. By Example 6.3, we have $y_n = P_n(x)$. Applying Lemma 6.5, we get

$$\|x\|^2 \geqslant \|P_n(x)\|^2 = \|\sum_{k=1}^{n} \langle x, v_k \rangle v_k\|^2 \underset{(A)}{=} \sum_{k=1}^{n} \|\langle x, v_k \rangle v_k\|^2 \underset{(B)}{=} \sum_{k=1}^{n} |\langle x, v_k \rangle|^2 \|v_k\|^2 \underset{(C)}{=} \sum_{k=1}^{n} |\langle x, v_k \rangle|^2$$

for every $x \in H$ and every $N \geqslant 1$. In (A) we used that the vectors v_n form an orthogonal system hence Pythagoras' theorem (Theorem 4.4) may be applied; (B) follows from the definition of norm, and the last equality (C) holds because $\|v_n\| = 1$ for all $n \geqslant 1$.

Since the LHS ($\|x\|^2$) is constant and the terms of the series $\sum_{n=1}^{\infty} |\langle x, v_n \rangle|^2$ are nonnegative, we conclude that the series $\sum_{n=1}^{\infty} |\langle x, v_n \rangle|^2$ converges and its sum is bounded by $\|x\|^2$ from above.

We now prove that $(y_n)_{n \geqslant 1}$ is Cauchy in H. Consider any $m > n \geqslant N$, and the definition of (y_n) in [1]. Then using that (D) follows from the Pythagoras' Theorem 4.4,

$$\|y_m - y_n\|^2 = \|\sum_{k=n+1}^{m} \langle x, v_k \rangle v_k\|^2 \underset{(D)}{=} \sum_{k=n+1}^{m} |\langle x, v_k \rangle|^2 \leqslant \sum_{k=N+1}^{\infty} |\langle x, v_k \rangle|^2 \underset{N \to \infty}{\to} 0,$$

as the series $\sum_{k=1}^{\infty} |\langle x, v_k \rangle|^2$ converges. Hence the sequence (y_n) is Cauchy, thus $y_n \to y$ for some $y \in H$. Moreover, as $y_n \in L$ for all $n \geqslant 1$ and L is closed, we have $y \in L$ by Lemma 2.19 (Sequence property).

From definition of y_n from [1] note that for $n \geqslant k$ we have $\langle y, v_k \rangle = \langle y, v_k \rangle = \langle x, v_k \rangle$. By Lemma 4.7 we have $\langle y, v_k \rangle = \lim_{n \to \infty} \langle y_n, v_k \rangle = \langle x, v_k \rangle$. Therefore, $\langle x - y, v_k \rangle = 0 \iff$

[3]Note that the linear subspaces L_n are different for different values of n, hence orthogonal projections P_n are different for different n as they project on different spaces.

$(x-y) \perp e_k$ for all $k \geqslant 1$, implying $(x-y) \in \mathcal{E}$. This implies $(x-y) \perp \langle \mathcal{E} \rangle$ by linearity, and $(x-y) \perp \overline{\langle \mathcal{E} \rangle}$ by Lemma 4.7. Thus $y = P_L(x)$ by definition of P_L.

Finally, as $y_n \to y$, we have by Theorem 2.8 (4) that $\|y\| = \lim_{n\to\infty} \|y_n\|$, hence $\|y\|^2 = \lim_{n\to\infty} \|y_n\|^2$. Applying again Pythagoras' Theorem 4.4 we get $\|y_n\|^2 = \sum_{k=1}^{n} |\langle x, v_k\rangle|^2$, hence $\|y\|^2 = \lim_{n\to\infty} \sum_{k=1}^{n} |\langle x, v_k\rangle|^2 = \sum_{k=1}^{\infty} |\langle x, v_k\rangle|^2$. This proves [2]. □

Corollary 6.7. *1. If H is a Hilbert space and $\{v_k\}$ is an infinite orthonormal sequence, then for any $x \in H$,* $\boxed{\sum_{k=1}^{\infty} |\langle x, v_k\rangle|^2 \leqslant \|x\|^2}$ *This inequality is referred to as **Bessel's inequality**.*

2. If H is a Hilbert space and $\{v_k\}$ is an infinite orthonormal sequence with dense span, then for any $x \in H$, $\boxed{\sum_{k=1}^{\infty} |\langle x, v_k\rangle|^2 = \|x\|^2}$ *This is referred to as **Parseval's identity**.*

Proof. 1. $\sum_{k=1}^{\infty} |\langle x, v_k\rangle|^2 = \|y\|^2 = \|P_L(x)\|^2 \leqslant \|x\|^2$, using Lemma 6.5.

2. The the linear span is dense, then $L = \overline{\langle \mathcal{E} \rangle} = H$, so $y = P_L(x) = P_H(x) = x$. Hence $\sum_{k=1}^{\infty} |\langle x, v_k\rangle|^2 = \|y\|^2 = \|x\|^2$. □

Examples (Bessel's inequality and Parseval's identity)

1. If $H = \ell^2$ and $e_n = (\underbrace{0, 0, \dots, 0}_{n-1 \text{ times}}, 1, 0, 0, \dots)$, then the collection $\{e_n : n \geqslant 1\}$ is orthonormal and its span is dense.

Hence for $x = (a_1, a_2, \dots) \in \ell^2$ we have, by Parseval's identity, $\sum_{k=1}^{\infty} |\langle x, e_k\rangle|^2 = \|x\|^2$. Indeed, we can check this: $|\langle x, e_n\rangle| = |a_n|$, so that $\sum_{n=1}^{\infty} |\langle x, e_n\rangle|^2 = \sum_{n=1}^{\infty} |a_n|^2 = \|x\|_2^2$ by the definition of $\|\cdot\|_2$.

2. However if we take only every second vector, $v_n = e_{2n} = (\underbrace{0, 0, \dots, 0}_{2n-1 \text{ times}}, 1, 0, 0, \dots)$, then the collection $\{v_n\}$ is orthonormal but its span is no longer dense in H.

Hence for $x = (a_1, a_2, \dots) \in \ell^2$ we have, by Bessel's inequality, $\sum_{k=1}^{\infty} |\langle x, v_k\rangle|^2 \leqslant \|x\|^2$. To check this: $|\langle x, v_n\rangle| = |a_{2n}|$, so that $\sum_{n=1}^{\infty} |\langle x, v_n\rangle|^2 = \sum_{n=1}^{\infty} |a_{2n}|^2 < \|x\|_2^2$ for some x, for example, $x = (1, 0, 0, 0, \dots)$.

Bounded linear functionals on Hilbert spaces

Theorem 6.8 (Riesz–Fréchet representation theorem)**.** *Let H be a Hilbert space; let $f \in H^*$. Then there exists a unique element $y \in H$ such that $f(x) = \langle x, y\rangle$ for each $x \in H$. Moreover, $\|f\|_{H^*} = \|y\|_H$.*

Proof. We will write $\|\cdot\|$ (without subscripts) for the norm of vectors from H.

The *uniqueness* claim is trivial because if $\langle x, y\rangle = \langle x, \widetilde{y}\rangle$ for each $x \in H$
then $\langle x, y - \widetilde{y}\rangle = 0$ for each $x \in H$, so by taking $x = y - \widetilde{y}$ it follows that $y = \widetilde{y}$ (we prove this in Question 16(a) Problem Sheet 2).

Existence:

• If $f \equiv 0$, then take $y = 0$: in that case $f(x) = 0 = \langle x, y\rangle$ for all $x \in H$ and $\|f\|_{H^*} = 0 = \|y\|$.

• Consider therefore the case when $f \not\equiv 0$. Note that $L = \ker f$ is a linear subspace of H:
- $0 \in \ker f$ as $f(0) = 0$ so that $\ker f \neq \emptyset$;
- if $x, y \in \ker f$, then $f(x) = f(y) = 0$ so that $f(\alpha x + y) = \alpha f(x) + f(y) = 0$ for all $\alpha \in \mathbb{F}$, hence $\alpha x + y \in \ker f$.

Since $\ker f$ is closed in H by Theorem 5.4, we conclude L is a closed linear subspace of H, so the orthogonal projection $P_L : H \to L$ is well-defined.

As $f \not\equiv 0$, we get $L = \ker f \neq H$. Let $v \in H \setminus L$ and $z := v - P_L(v)$. Then $z \perp L$ and $z \notin L$, as $v \notin L$ and $P_L(v) \in L$.

As $z \notin L = \ker f$, we have $f(z) \neq 0$. Define $w = z/f(z)$, then we have $w \perp L$ and $f(w) = 1$, hence $w \neq 0_H$. For each $x \in H$, the vector $x - f(x)w$ belongs to $\ker f = L$. This is because $f\Big(x - f(x)w\Big) = f(x) - f(x)f(w) = 0$. Recall $w \perp L$, hence $w \perp (x - f(x)w)$, i.e. $\langle x - f(x)w, w\rangle = 0$. From linearity of the inner product we conclude that

$$\langle x, w\rangle = \langle f(x)w, w\rangle = f(x)\langle w, w\rangle = f(x)\|w\|^2.$$

As $w \neq 0_H$, we have $\|w\| \neq 0$, so divide both sides by $\|w\|^2$ to get $\langle x, y\rangle = f(x)$ for all $x \in H$, where $y = w/\|w\|^2$. This proves the existence of such y.

To prove $\|f\|_{H^*} = \|y\|$, note $|f(x)| = |\langle x, y\rangle| \leqslant \|x\|\|y\|$ by the Cauchy–Schwarz inequality, Theorem 4.5, and consequently $\|f\|_{H^*} \leqslant \|y\|$. By Lemma 5.2

$$\|y\|^2 = |\langle y, y\rangle| = |f(y)| \leqslant \|f\|_{H^*}\|y\|, \text{ so } \|f\|_{H^*} \geqslant \|y\|.$$

The two inequalities for $\|f\|_{H^*}$ imply $\|f\|_{H^*} = \|y\|$. □

QUICK POLLS: 6. HILBERT SPACES See Power Point presentation 6, question 7.

We therefore see that the space of bounded linear functionals on a Hilbert space is, in some sense, "identical" to the space H itself: any $f \in H^*$ is identified with y as per Theorem 6.8. This can be written as $H^* = H$ for a Hilbert space H over $\mathbb{R}$. Thus $(\ell^2)^* = \ell^2$.

§7 Linear operators

Definition (Linear operator)

Let X and Y be vector spaces over the same field $\mathbb{F}$, and $U : X \to Y$ satisfy the following properties:

$$U(\alpha x + y) = \alpha U(x) + U(y)$$

for all $x, y \in X$ and all scalars $\alpha \in \mathbb{F}$. Then U is called a *linear operator*.

Definition (Bounded linear operator)

If $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ are normed spaces and $U : X \to Y$ is a linear operator, we say that U is a *bounded linear operator* if there exists $0 \leqslant C < \infty$ such that $\|U(x)\|_Y \leqslant C\|x\|_X$ for all $x \in X$. We write $\mathcal{B}(X, Y)$ for the space of all bounded linear operators from X to Y.

Theorem 7.1. *Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces. Then*
(1) $\mathcal{B}(X, Y)$ is a vector space over $\mathbb{F}$.
(2) $\mathcal{B}(X, Y)$ is a normed space with the operator norm:

$$\|U\| = \inf\{C \geqslant 0 \colon \|U(x)\|_Y \leqslant C\|x\|_X \quad \forall x \in X\} = \sup_{\|x\|_X \leqslant 1} \{\|U(x)\|_Y\}.$$

(3) $\|U(x)\|_Y \leqslant \|U\|\|x\|_X$ for all $x \in X$.
(4) $U \in \mathcal{B}(X, Y)$ if and only if $U : X \to Y$ is continuous at every $x \in X$ if and only if $U : X \to Y$ is continuous at $x = 0$.
(5) If $(Y, \|\cdot\|_Y) = (\mathbb{F}, |\cdot|)$, then $\mathcal{B}(X, Y) = X^*$.
(6) If $X = Y = H$ is a Hilbert space and $L \neq \{0\}$ is a closed linear subspace, then $P_L \in \mathcal{B}(H, H)$ and $\|P_L\| = 1$.

Proof. (1) The proof is analogous to the proof of Lemma 5.1.
(2) and (3) The proof is analogous to the solution of Q24 from PS2 (it is a norm), Proposition 5.3 and Lemma 5.2.
(4) The proof is analogous to the proof of Theorem 5.4. Note that there is no equivalent condition involving $\ker U$!
(5) By the definition of X^*.
(6) By Theorems 6.1 and 6.2, and the definition of P_L, for every $x \in H$ the vector $P_L(x)$ is the unique $y \in L$ such that $(x - y) \in L^\perp$. So if $x_1, x_2 \in H$ and $\alpha \in \mathbb{F}$, then let $y_1 = P_L(x_1)$, $y_2 = P_L(x_2)$, then $\alpha y_1 + y_2 \in L$ since L is a linear subspace, and

$$(\alpha x_1 + x_2) - (\alpha y_1 + y_2) = \alpha(x_1 - y_1) + (x_2 - y_2) \in L^\perp, \text{ since } L^\perp \text{ is a linear subspace.}$$

Hence $P_L(\alpha x_1 + x_2) = \alpha y_1 + y_2$, so P_L is a linear operator.

From Lemma 6.5, $\|P_L(x)\|_H \leqslant \|x\|_H$ for all $x \in H$, so P_L is a bounded linear operator and $\|P_L\| \leqslant 1$. Take any $v \in L \setminus \{0_H\}$, then $P_L(v) = v$, so using (3) of the present theorem, $\|v\|_H = \|P_L(v)\|_H \leqslant \|P_L\|\|v\|_H$, implying $\|P_L\| \geqslant 1$, using $\|v\|_H \neq 0$. □

Example (Linear operators, bounded and unbounded)

1. Let $U : (\ell^p, \|\cdot\|_p) \to (\ell^p, \|\cdot\|_p)$ be the *shift* operator, defined as $U(a_1, a_2, a_3, \dots) = (a_2, a_3, a_4, \dots)$.

Then U is a linear operator as $U(\alpha x + y) = U(\alpha(a_n)_{n\geqslant 1} + (b_n)_{n\geqslant 1}) = U((\alpha a_n + b_n)_{n\geqslant 1}) = (\alpha a_n + b_n)_{n\geqslant 2}) = \alpha(a_n)_{n\geqslant 2} + (b_n)_{n\geqslant 2} = \alpha U(x) + U(y)$

and

$\|U(x)\|_p = \left(\sum_{i=2}^{\infty} |a_i|^p\right)^{1/p} \leqslant \left(\sum_{i=1}^{\infty} |a_i|^p\right)^{1/p} = \|x\|_p$ if $1 \leqslant p < \infty$;

$\|U(x)\|_\infty = \sup_{i\geqslant 2} |a_i| \leqslant \sup_{i\geqslant 1} |a_i| = \|x\|_\infty$.

Hence U is a bounded linear operator.

Exercise: check $\|U\| = 1$.

2. Let $V : \ell^2 \to \ell^1$ be defined by $V(a_1, a_2, a_3, \dots) = (a_1^2, a_2^2, a_3^2, \dots)$.

Is V well defined? Is it a linear operator?

QUICK POLLS: 7. LINEAR OPERATORS See Power Point presentation 7, questions 1–4.

The following examples were not covered in lectures and are given here in addition (non-examinable).

3. Let $\mathcal{P} = \mathcal{P}[0,1]$ denote the subspace of $(C[0,1], \|\cdot\|_\infty)$ consisting of restrictions to $[0,1]$ of polynomials with real coefficients. Define $W : \mathcal{P} \to \mathcal{P}$ by $W(p) = p'$, the derivative of p.

It is easy to see that W is a linear operator. But is it bounded?

No, this linear operator is not bounded.

Assume $\exists C \geqslant 0$ such that $\|W(p)\|_\infty \leqslant C\|p\|_\infty$ for every $p \in \mathcal{P}$. Let $n > C + 1$ and $p(t) = t^n$. Then $\|p\|_\infty = |p(1)| = 1$ and $\|W(p)\|_\infty = \|p'\|_\infty = |p'(1)|_\infty = n$. Then $n \leqslant C$, a contradiction.

It is interesting to see that the kernel of W is closed. Compare this with the Theorem 5.4.

Indeed, $p \in \ker W$ if and only if $p'(t) = 0$ for all $t \in [0,1]$, which is equivalent to p being a constant c on $[0,1]$, hence on the whole of $\mathbb{R}$ ($p(t) - c$ will have infinitely many roots which means it is the zero polynomial).

So $\ker W = \{\text{constant polynomials}\}$. It remains to verify that $\ker W$ satisfies the Sequence property, Lemma 2.19. Indeed, if $p_n(t) \equiv c_n \in \ker W$ converges to p in $(\mathcal{P}, \|\cdot\|_\infty)$, then for each $t_0 \in [0,1]$

$$|p(t_0) - p_n(t_0)| \leqslant \|p - p_n\|_\infty \to 0$$

implies that $p(t_0) = \lim c_n$. Thus, p is also a constant polynomial.

4. Let $(X, \|\cdot\|) = (C[0,1], \|\cdot\|_\infty)$. Define, for each $f \in X$, the function $g = I(f)$ by

$$g(t) = \int_0^t f(y)dy.$$

Since $f(y)$ is continuous, the integral above converges. Moreover, g is a continuous function since,

$$|g(t) - g(t')| = \left|\int_{t'}^t f(y)dy\right| \leqslant \int_{t'}^t |f(y)|dy \leqslant |t - t'| \cdot \|f\|_\infty \to 0 \text{ as } t' \to t.$$

The operator $I : X \to X$ is called the **Integral operator**. It is easy to check that this is a linear operator. Is it bounded? For every $t \in [0,1]$,

$$|I(f)(t)| \leqslant \int_0^t |f(y)|dy \leqslant t\|f\|_\infty \leqslant \|f\|_\infty,$$

so $\|I(f)\|_\infty \leqslant \|f\|_\infty$ for all $f \in X$. We conclude that I is a bounded linear operator and $\|I\| \leqslant 1$. Take $f_0 \equiv 1$, then $\|f_0\|_\infty = 1$, so

$$\|I\| \geqslant \|I(f_0)\|_\infty \geqslant |I(f_0)(1)| = \int_0^1 1dy = 1,$$

this proves $\|I\| = 1$.

If we considered $C[a,b]$ instead of $C[0,1]$, we would get $\|I\| = b - a$.

Hilbert adjoint

Let H_1, H_2 be Hilbert spaces over the same $\mathbb{F}$. Let $U : H_1 \to H_2$ be a bounded linear operator. Fix $y \in H_2$ and define $f(x) = f_y(x) = \langle U(x), y\rangle_{H_2}$, for each $x \in H_1$. Then

- $f : H_1 \to \mathbb{F}$ is well-defined,
- f is a linear functional, as
  $f(ax_1 + x_2) = \langle U(ax_1 + x_2), y\rangle_{H_2} = a\langle U(x_1), y\rangle_{H_2} + \langle U(x_2), y\rangle_{H_2}$,
- f is a bounded linear functional, as
  $|f(x)| = |\langle U(x), y\rangle_{H_2}| \leqslant \|U(x)\|_{H_2}\|y\|_{H_2} \leqslant \|U\|\|x\|_{H_1}\|y\|_{H_2} = \Big(\|U\|\|y\|_{H_2}\Big)\|x\|_{H_1}$,
- and $\|f\|_{H_1^*} \leqslant \|U\|\|y\|_{H_2}$.

By Riesz-Fréchet Representation Theorem 6.8 there exists a unique $z \in H_1$ such that

$$\langle U(x), y\rangle_{H_2} = f(x) = \langle x, z\rangle_{H_1} \text{ for all } x \in H_1.$$

Define $U' : H_2 \to H_1$ by $U'(y) := z$ as above, then

$$\langle U(x), y\rangle_{H_2} = \langle x, U'(y)\rangle_{H_1}, \text{ for all } x \in H_1, y \in H_2.$$

Note that

- $U' : H_2 \to H_1$ is well-defined,
- U' is a linear operator, as for all $x \in H_1$
  $\langle x, U'(ay_1+y_2)\rangle_{H_1} = \langle U(x), ay_1+y_2\rangle_{H_2} = \bar{a}\langle U(x), y_1\rangle_{H_2}+\langle U(x), y_2\rangle_{H_2} = \langle x, aU'(y_1)\rangle_{H_1}+\langle x, U'(y_2)\rangle_{H_1} = \langle x, aU'(y_1) + U(y_2)\rangle_{H_1}$, so $U'(ay_1 + y_2) = aU'(y_1) + U'(y_2)$ by Question 17a.
- U' is a bounded linear operator, as for all $y \in H_2$
  $\|U'(y)\|_{H_1} = \|z\|_{H_1} = \|f\|_{H_1^*} \leqslant \|U\|\|y\|_{H_2}$.
- The above implies $\|U'\| \leqslant \|U\|$. To show $\|U'\| = \|U\|$, let $y = U(x)$, then

$$\|y\|_{H_2}^2 = \|U(x)\|_{H_2}^2 = |\langle U(x), y\rangle_{H_2}| = |\langle x, U'(y)\rangle_{H_1}| \leqslant \|x\|_{H_1}\|U'\|\|y\|_{H_2},$$

  so $\|U(x)\|_{H_2} = \|y\|_{H_2} \leqslant \|U'\|\|x\|_{H_1}$, for all $x \in H_1$. This implies $\|U\| \leqslant \|U'\|$.

This bounded linear operator U' is called the Hilbert adjoint and generalises the transpose matrix.

§8 Appendix A: Proof of Theorem 3.3

Theorem 3.3. *For all* $1 \leqslant p \leqslant +\infty$, $(\ell^p, \|\cdot\|_p)$ *is a Banach space.*

Proof. Case 1: $p = +\infty$. This is done in the Problem Sheet 1, Question 15.

Case 2: $1 \leqslant p < +\infty$. To prove the claim, we need to consider an arbitrary Cauchy sequence of vectors $v_n \in \ell^p$ and to show that that it converges.

Let $(v_n)_{n\geqslant 1}$ be a Cauchy sequence in ℓ^p, and write

$$v_n = (a_{1,n}, a_{2,n}, a_{3,n}, \dots) = (a_{k,n})_{k\in\mathbb{N}}.$$

For each $\varepsilon > 0$, let $N_\varepsilon \in \mathbb{N}$ be such that $\|v_n - v_m\|_p < \varepsilon$ whenever $n, m \geqslant N_\varepsilon$, i.e.

$$\sum_{i=1}^{\infty} |a_{i,n} - a_{i,m}|^p = \big(\|v_n - v_m\|_p\big)^p < \varepsilon^p \qquad \text{for all } n, m \geqslant N_\varepsilon. \tag{8.1}$$

By Lemma 2.12 we also have that for each fixed $k \geqslant 1$

$$|a_{k,n} - a_{k,m}| \leqslant \|v_n - v_m\| < \varepsilon \qquad \text{for all } n, m \geqslant N_\varepsilon.$$

The latter implies that for each fixed $k \in \mathbb{N}$, we have that $(a_{k,n})_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$. By the completeness axiom, for every fixed $k \in \mathbb{N}$ there exists $y_k \in \mathbb{R}$ such that $a_{k,n} \underset{n\to\infty}{\to} y_k$.

We let $y := (y_k)_{k\geqslant 1}$ and claim that $y \in \ell^p$ and moreover $\|v_n - y\|_p \to 0$.

Firstly, (8.1) implies that for every fixed $T \geqslant 1$ we have that for all $n, m \geqslant N_\varepsilon$

$$\sum_{i=1}^{T} |a_{i,n} - a_{i,m}|^p < \varepsilon^p. \tag{8.2}$$

As we have a finite sum in the left hand side of (8.2) we can let $m \to \infty$ to get that for all $n \geqslant N_\varepsilon$

$$\sum_{i=1}^{T} |a_{i,n} - y_s|^p \leqslant \varepsilon^p.$$

Finally in the inequality above let $T \to \infty$. We get for all $n \geqslant N_\varepsilon$

$$\sum_{i=1}^{\infty} |a_{i,n} - y_s|^p \leqslant \varepsilon^p.$$

This means that for all $n \geqslant N_\varepsilon$ the vectors $(v_n - y) \in \ell^p$ and $\|v_n - y\|_p \leqslant \varepsilon$.

This implies that $y = v_{N_1} - (v_{N_1} - y) \in \ell^p$ (using that ℓ^p is a linear space, see Corollary 2.4) and $\|v_n - y\|_p \to 0$.

Hence we proved the sequence (v_n) converges in ℓ^p. □

§9 Appendix B: Proof of Theorem 3.8

The sequence $(g_n)_{n\geqslant 1}$ of functions from $C[0,2]$ defined in Example 3.7 by

$$g_n(t) = \begin{cases} t^n, & \text{if } 0 \leqslant t \leqslant 1; \\ 1, & \text{if } 1 \leqslant t \leqslant 2. \end{cases}$$

does not have a limit in $(C[0,2], \|\cdot\|_1)$.

Proof of Theorem 3.8. We show that for every $h \in C[0,2]$ we have $\|g_n - h\|_1 \not\to 0$. This implies that the sequence g_n does not converge in $(C[0,2], \|\cdot\|_1)$, thus the space $(C[0,2], \|\cdot\|_1)$ is not a Banach space, as we have shown in §3 that (g_n) is Cauchy in $(C[0,2], \|\cdot\|_1)$.

Let $h \in C[0,2]$ be any continuous function, denote $d = h(1)$.

If $d \neq 1$, by continuity of h there exists $r \in (0,1)$ such that $|h(t)-d| = |h(t)-h(1)| < |1-d|/2$ for all $t \in (1-r, 1+r)$, which implies, in particular,

$$|h(t) - 1| \geqslant |1-d|/2 \text{ for all } t \in [1, 1+r].$$

We have:

$$\|g_n - h\|_1 = \|h - g_n\|_1 \geqslant \int_1^{1+r} |h(t) - g_n(t)|dt = \int_1^{1+r} |h(t) - 1|dt \geqslant \frac{|1-d|}{2} r,$$

which is a positive constant independent of n. Thus $\|g_n - h\|_1$ does not tend to 0 as $n \to \infty$.

If $d = 1$, then by continuity of h there exists $r \in (0,1)$ such that $|h(t)-d| = |h(t)-h(1)| < 0.1$ for all $t \in (1-r, 1+r)$, which implies, in particular,

$$h(t) \geqslant 0.9 \qquad \text{for all } t \in [1-r, 1].$$

As $\left(1-\frac{r}{2}\right)^n \to 0$, choose N such that $\left(1-\frac{r}{2}\right)^N < 0.4$, thus $t^n < 0.4$ for all $t \in \left[1-r, 1-\frac{r}{2}\right]$ and all $n \geqslant N$. This implies that

$$|h(t) - g_n(t)| = |h(t) - t^n| > 1/2 \qquad \text{for all } t \in \left[1-r, 1-\tfrac{r}{2}\right].$$

Finally, we conclude that for all $n \geqslant N$

$$\|g_n - h\|_1 = \|h - g_n\|_1 \geqslant \int_{1-r}^{1-r/2} |h(t) - g_n(t)|dt \geqslant \tfrac{r}{2} \times \tfrac{1}{2} = \tfrac{r}{4},$$

which is a positive constant independent of n. Therefore, $\|g_n - h\|_1$ does not tend to 0 as $n \to \infty$. □

Remark. Thus we have that the sequence of continuous functions (g_n) is Cauchy in $(C[0,2], \|\cdot\|_1)$ and has no limit in this space. This proves that $(C[0,2], \|\cdot\|_1)$ is not a Banach space, which is essentially Theorem 3.8 (with $a = 0$ and $b = 2$).

Note that moreover we can prove the following theorem. We refer to this theorem in Example [3] in the beginning of §6, to argue that the space $(C[a,b], \|\cdot\|_2)$ is not a Hilbert space. In fact, we prove a more general statement.

Theorem 9.1. *Let* $p \in [1,\infty)$ *be arbitrary. Then* $(C[a,b], \|\cdot\|_p)$ *is not a Banach space, where* $\|\cdot\|_p$ *is defined as follows:*

$$\|f\|_p = \Big(\int_a^b |f(t)|^p \, dt\Big)^{1/p}$$

for any $f \in C[a,b]$.

Proof. Without loss of generality assume $a = 0$ and $b = 2$. Consider the sequence of continuous functions g_n, as defined above. We have $\big(\|g_n - g_m\|_p\big)^p = \int_0^2 |g_n(t) - g_m(t)|^p dt = \int_0^1 |t^n - t^m|^p dt \leqslant \int_0^1 |t^n - t^m| dt = \Big|\frac{1}{n} - \frac{1}{m}\Big|$, so the same sequence (g_n) is Cauchy in $(C[a,b], \|\cdot\|_p)$.

Assuming that $h \in C[a,b]$ is arbitrary, we consider two cases, $h(1) \neq 1$ and $h(1) = 1$ similarly to above and arrive in a similar way to inequalities

$$\begin{cases} \big(\|g_n - h\|_p\big)^p \geqslant \Big(\frac{|1-d|}{2}\Big)^p r, & \text{if } d \neq 1 \text{ and for all } n \geqslant 1; \\ \big(\|g_n - h\|_p\big)^p \geqslant \frac{r}{2}\big(\frac{1}{2}\big)^p, & \text{if } d = 1 \text{ and } n \text{ is sufficiently large.} \end{cases}$$

This means that $\big(\|g_n - h\|_p\big)^p \not\to 0$, so that $\|g_n - h\|_p \not\to 0$. This implies that this Cauchy sequence (g_n) does not have a limit in $(C[0,2], \|\cdot\|_p)$. □

§10 Appendix C: Proof of Young's inequality

Here we prove inequality (4.2) from the proof of Hölder's inequality, Theorem 4.2: whenever $s, t \geqslant 0$ and $p, q > 1$ are such that $\frac{1}{p} + \frac{1}{q} = 1$, it holds

$$st \leqslant \frac{s^p}{p} + \frac{t^q}{q}. \tag{10.1}$$

Define $a = s^p$ and $b = t^q$, so that $s = a^{1/p}$ and $t = b^{1/q}$. We then need to prove that

$$a^{1/p} b^{1/q} \leqslant \frac{a}{p} + \frac{b}{q} \tag{10.2}$$

for all $a, b \in [0, \infty)$. Assume without loss of generality $a \leqslant b$. The inequality is obvious for $a = 0$ and for $a = b$, so we may further assume $b > a > 0$.

Let $f(t) = t^{1/q}$. Since $b > a > 0$, the function f is continuous on $[a, b]$ and is differentiable on (a, b). Then by the Mean Value Theorem,

$$b^{1/q} - a^{1/q} = f(b) - f(a) = f'(c)(b - a) = \frac{c^{\frac{1}{q}-1}}{q}(b - a) = \frac{c^{-1/p}(b - a)}{q}$$

for some $c \in (a, b)$. Here we used $\frac{1}{q} - 1 = -\frac{1}{p}$. As $c > a$, we have $c^{-1/p} < a^{-1/p}$ (and in writing so, we use $a > 0$), so it follows

$$b^{1/q} - a^{1/q} \leqslant \frac{a^{-1/p}(b - a)}{q},$$

which after multiplying by $a^{1/p}$ gives

$$a^{1/p} b^{1/q} - a \leqslant (b - a)/q.$$

This is equivalent to

$$a^{1/p} b^{1/q} \leqslant \frac{b}{q} + (a - \frac{a}{q}) = \frac{a}{p} + \frac{b}{q}.$$

This proves (10.2), which, in turn, implies (10.1), that is, inequality (4.2).

§11 Appendix D: Parallelogram Law

In this section we prove Theorem 4.9.

Theorem 11.1 (Theorem 4.9)**.** *Let* $(X, \|\cdot\|)$ *be a normed vector space such that the Parallelogram law* (4.6) *holds for any* $x, y \in X$*. Then there exists an inner product* $\langle\cdot,\cdot\rangle : X \times X \to \mathbb{F}$ *which induces the norm* $\|\cdot\|$*; i.e.* $\langle x, x\rangle^{1/2} = \|x\|$ *for each* $x \in X$*.*

Proof. We include the proof for the case $\mathbb{F} = \mathbb{R}$. Assume the parallelogram law (4.6) is satisfied for all $x, y \in X$; define $\langle\cdot,\cdot\rangle : X \times X \to \mathbb{R}$ by[4]

$$\langle x, y\rangle := \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2). \tag{11.1}$$

It is clear that $\langle x, x\rangle = \|x\|^2$ for each $x \in X$ by the 2nd property of $\|\cdot\|$. Hence the 1st property of $\langle\cdot,\cdot\rangle$ is satisfied. It is also clear that $\langle\cdot,\cdot\rangle$ is symmetric using the 2nd property of $\|\cdot\|$: $\langle y, x\rangle = \langle x, y\rangle$. It remains to verify the linearity condition. To this end, the proof follows the following steps. For all $x, y, z \in X$

1. $\langle 0, z\rangle = \frac{1}{2}(\|0+z\|^2 - \|0-z\|^2) = \frac{1}{2}(\|0+z\|^2 - \|z-0\|^2) = 0.$
2. $2\left\langle \frac{x+y}{2}, z\right\rangle = \langle x, z\rangle + \langle y, z\rangle$:

$$\begin{aligned}
2\left\langle \frac{x+y}{2}, z\right\rangle &= \frac{1}{2}\left(\left\|\frac{x+y}{2}+z\right\|^2 - \left\|\frac{x+y}{2}-z\right\|^2\right) \\
&= \frac{1}{2}\left(\left\|\frac{x+y}{2}+z\right\|^2 + \left\|\frac{x-y}{2}\right\|^2\right) - \frac{1}{2}\left(\left\|\frac{x+y}{2}-z\right\|^2 + \left\|\frac{x-y}{2}\right\|^2\right) \\
&\underset{*}{=} \frac{1}{4}\left(\left\|\frac{x+y}{2}+z+\frac{x-y}{2}\right\|^2 + \left\|\frac{x+y}{2}+z-\frac{x-y}{2}\right\|^2\right. \\
&\qquad \left. - \left\|\frac{x+y}{2}-z+\frac{x-y}{2}\right\|^2 - \left\|\frac{x+y}{2}-z-\frac{x-y}{2}\right\|^2\right) \\
&= \frac{1}{4}(\|x+z\|^2 + \|y+z\|^2 - \|x-z\|^2 - \|y-z\|^2),
\end{aligned}$$

where equality marked * follows from the parallelogram law (4.6) which we assumed holds for all pairs of vectors from X.

Applying again (11.1), we obtain that the latter is equal to $\langle x, z\rangle + \langle y, z\rangle$, hence

$$2\left\langle \frac{x+y}{2}, z\right\rangle = \langle x, z\rangle + \langle y, z\rangle \tag{11.2}$$

for all $x, y, z \in X$.

[4]The proof for $\mathbb{F} = \mathbb{C}$ proceeds in a similar way under the definition, for all $x, y \in X$

$$\langle x, y\rangle := \frac{1}{4}\Big[(\|x+y\|^2 - \|x-y\|^2) + i(\|x+iy\|^2 - \|x-iy\|^2)\Big].$$

3. Taking $y = 0$ in (11.2) gives, for all $x, z \in X$,

$$2\langle x/2, z\rangle = \langle x, z\rangle, \quad \text{i.e.,} \quad 2\langle u, z\rangle = \langle 2u, z\rangle$$

for all $u, z \in X$.

4. Using (11.2) again and step 3, we obtain, for all $x, y, z \in X$,

$$\langle x, z\rangle + \langle y, z\rangle = 2\langle (x+y)/2, z\rangle = \langle x+y, z\rangle. \tag{11.3}$$

5. Note that (11.3) immediately implies ($u = x + y$)

$$\langle u, z\rangle - \langle y, z\rangle = \langle u - y, z\rangle \tag{11.4}$$

for all $u, y, z \in X$, so if we put $u = 0$ and use Step 1, then

$$-\langle y, z\rangle = \langle -y, z\rangle \tag{11.5}$$

for all $y, z \in X$.

6. Our aim now is to verify that

$$\langle \alpha x, z\rangle = \alpha\langle x, z\rangle \tag{11.6}$$

for all $\alpha \in \mathbb{R}$ and all $x, y \in X$. We will do this first for α in $\mathbb{N}$, then in $\mathbb{Z}$, then in $\mathbb{Q}$ and finally in $\mathbb{R}$.

Start by noticing (11.6) is satisfied for $\alpha = 2$ from Step 3. Hence, by induction and (11.3), we get

$$m\langle x, z\rangle = \langle mx, z\rangle \tag{11.7}$$

for all $m \in \mathbb{N}$ and all $x, z \in X$, i.e. (11.6) is satisfied for $\alpha \in \mathbb{N}$. We can immediately extend (11.7) to all $m \in \mathbb{Z}$ using (11.5) and step 1.

Note that (11.7) implies $n\langle x/n, z\rangle = \langle x, z\rangle$ for all $n \in \mathbb{N}$ and all $x, z \in X$, in other words,

$$\langle \frac{1}{n}x, z\rangle = \frac{1}{n}\langle x, z\rangle.$$

Hence, applying (11.7),

$$\frac{m}{n}\langle x, z\rangle = \Big\langle \frac{m}{n}x, z\Big\rangle \tag{11.8}$$

for all $m \in \mathbb{Z}, n \in \mathbb{N}$ and all $x, z \in X$, i.e. (11.6) is satisfied for all $\alpha \in \mathbb{Q}$.

We now claim that, for fixed $x, z \in X$ and $\alpha_n \to \alpha$ we have

$$\langle \alpha_n x, z\rangle \to \langle \alpha x, z\rangle.$$

Indeed, by (11.1) and continuity of the norm and linear combinations from Theorem 2.8

$$\langle \alpha_n x, z\rangle = \frac{1}{4}\big(\|\alpha_n x + z\|^2 - \|\alpha_n x - z\|^2\big) \to \frac{1}{4}\big(\|\alpha x + z\|^2 - \|\alpha x - z\|^2\big) = \langle \alpha x, z\rangle.$$

On the other hand, if $\alpha_n \in \mathbb{Q}$, then we have, using (11.6) for rational values, $\langle \alpha_n x, z\rangle = \alpha_n \langle x, z\rangle \to \alpha\langle x, z\rangle$, hence (11.6) is satisfied for any $\alpha \in \mathbb{R}$.

This completes the proof of Theorem 4.9 as linearity of $\langle \cdot, \cdot \rangle$ follows when we add formula (11.3) from Step 4:

$$\langle \alpha x + y, z\rangle = \langle \alpha x, z\rangle + \langle y, z\rangle = \alpha\langle x, z\rangle + \langle y, z\rangle.$$

□

§12 Appendix E: Calculate norm of projection

Here we show how to calculate a norm of projection.

Example (Calculating $\|Px\|$)

Let $H = \ell^2$ and $L = \langle 4e_1 - e_5, e_3 \rangle$, where e_k is the standard vector $(\underbrace{0, \dots, 0}_{k-1}, 1, 0, 0, \dots)$. Let $x = (2, 7, 8, 5, -9, 3, 7, -2, 0, 0, 0, \dots)$.

Let $P_L : H \to L$ be the orthogonal projection. Find $\|P_L(x)\|_2$.

We note first that $L = \langle u_1, u_2 \rangle$, where $u_1 = \frac{4e_1 - e_5}{\|4e_1 - e_5\|_2}$ and $u_2 = e_3$, and $\{u_1, u_2\}$ is an orthonormal collection of vectors (this is because $e_1 \perp e_3$ and $e_5 \perp e_3$).

Hence by Example 6.3 we have $P_L(x) = \langle x, u_1 \rangle u_1 + \langle x, u_2 \rangle u_2$.

We now find $\langle x, u_1 \rangle u_1$ and $\langle x, u_2 \rangle u_2$.

$\langle x, u_1 \rangle = \langle x, \frac{4e_1 - e_5}{\|4e_1 - e_5\|} \rangle$. As $4e_1 - e_5 = (4, 0, 0, 0, -1, 0, 0, \dots)$, we have

$\|4e_1 - e_5\| = \|4e_1 - e_5\|_2 = \sqrt{4^2 + 0^2 + 0^2 + 0^2 + |-1|^2 + 0^2 + 0^2 + 0^2 + \dots} = \sqrt{17}$.

Thus $\langle x, u_1 \rangle = \langle x, \frac{4e_1 - e_5}{\sqrt{17}} \rangle = \frac{4}{\sqrt{17}} \langle x, e_1 \rangle - \frac{1}{\sqrt{17}} \langle x, e_5 \rangle = \frac{4}{\sqrt{17}} \times 2 - \frac{1}{\sqrt{17}} \times (-9) = \sqrt{17}$.

We conclude $\langle x, u_1 \rangle u_1 = \sqrt{17} \frac{4e_1 - e_5}{\sqrt{17}} = 4e_1 - e_5$.

The second term is easy to compute: $\langle x, u_2 \rangle u_2 = \langle x, e_3 \rangle e_3 = 8e_3$.

We get: $P_L(x) = 4e_1 - e_5 + 8e_3 = (4, 0, 8, 0, -1, 0, 0, 0, \dots)$, hence

$\|P_L(x)\| = \sqrt{4^2 + 0^2 + 8^2 + 0^2 + |-1|^2 + 0^2 + 0^2 + 0^2 \dots} = \sqrt{81} = 9$.

§13 Appendix F: Density of span

We use Theorem 6.4 in order to decide whether the span of a subset of ℓ^2 is dense in ℓ^2.

1. Let $M = \{e_1, e_2, \dots, e_n, \dots\}$, where $e_n \in \ell^2$ is the vector which has all coordinates equal to 0 except the one which stands on the nth place and equals 1. We show that $\langle M\rangle$ is dense in ℓ^2.

By Theorem 6.4, to show $\langle M\rangle$ is dense in $H = \ell^2$ is equivalent to show that $M^\perp = \{0\}$.

$x \in M\perp \iff x \perp e_n\ \forall n \geqslant 1 \iff \langle x, e_n\rangle = 0\ \forall n \geqslant 1$.

Let $x = (a_1, a_2, a_3, \dots)$. Then $\langle x, e_n\rangle = a_n$, so $x \in M^\perp \iff a_n = 0\ \forall n \geqslant 1$.

$a_n = 0\ \forall n \geqslant 1$ means that $x = (0, 0, 0, \dots)$. Hence we proved $M^\perp = \{0\}$.

By Theorem 6.4 this implies the linear span of M, the space $\langle M\rangle$, is dense in ℓ^2.

Note that $\langle e_1, e_2, \dots\rangle = c_{00}$ (see discussion in the end of Section §1). Hence we proved $\overline{c_{00}} = \overline{\langle M\rangle} = H = \ell^2$, or, equivalently, c_{00} is dense in ℓ^2.

2. Consider $M = \{e_k + \frac{1}{2}e_{k+1} : k \geqslant 1\}$. Is $\langle M\rangle$ dense in ℓ^2?

To answer this question, let us calculate $M^\perp$:

$x \in M^\perp \iff x \perp M \iff \langle x, e_k + \frac{1}{2}e_{k+1}\rangle = 0\ \forall k \geqslant 1$. Let $x = (a_1, a_2, a_3, \dots)$. Then we get $\langle x, e_k + \frac{1}{2}e_{k+1}\rangle = 0\ \forall k \geqslant 1 \iff a_k + \frac{1}{2}a_{k+1} = 0\ \forall k \geqslant 1 \iff a_{k+1} = -2a_k\ \forall k \geqslant 1$.

Let $a_1 = a$, then $a_2 = -2a$, $a_3 = 4a$, $a_4 = -8a$, $\dots$, $a_n = \pm 2^{n-1}a$, and thus

$$x = (a, -2a, 4a, -8a, \dots).$$

It is clear that if $a \neq 0$, then $\sum_{k=1}^{\infty} |x_k|^2 = \infty$ which contradicts $x \in \ell^2$. Hence $a = 0$ which implies $x = 0$, thus $M^\perp = \{0\}$.

By Theorem 6.4 this implies the linear span of M, the space $\langle M\rangle$, is dense in ℓ^2.

3. Consider now $M = \{e_k + 2e_{k+1} : k \geqslant 1\}$. Is $\langle M\rangle$ dense in ℓ^2?

Will the result change is 1/2 is replaced with 2?

To answer this question, let us calculate $M^\perp$:

$x \in M^\perp \iff x \perp M \iff \langle x, e_k + 2e_{k+1}\rangle = 0\ \forall k \geqslant 1$. Denote coordinates of x by $(a_1, a_2, a_3, \dots)$. Then we get $\langle x, e_k + 2e_{k+1}\rangle = 0\ \forall k \geqslant 1 \iff a_k + 2a_{k+1} = 0\ \forall k \geqslant 1 \iff$

$$a_{k+1} = -\frac{1}{2}a_k \quad \forall k \geqslant 1. \tag{*}$$

We see then that the (non-zero) vector $v = (1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \dots) \in \ell^2$ satisfies the condition $(*)$, and hence $v \in M^\perp$ which implies $M^\perp \neq \{0\}$.

By Theorem 6.4 this implies the linear span of M, the space $\langle M\rangle$, is NOT dense in ℓ^2.